

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406416
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Bondich; Artem et al.

Avatar based ideogram generation

Abstract

Systems, devices, media, and methods are presented for generating ideograms from a set of images received in an image stream. The systems and methods detect at least a portion of a face within the image and identify a set of facial landmarks within the portion of the face. The systems and methods determine one or more characteristics representing the portion of the face, in response to detecting the portion of the face. Based on the one or more characteristics and the set of facial landmarks, the systems and methods generate a representation of a face. The systems and methods position one or more graphical elements proximate to the graphical model of the face and generate an ideogram from the graphical model and the one or more graphical elements.

Inventors: Bondich; Artem (Marina del Rey, CA), Maltsev; Vladimir (Playa Vista, CA)

Applicant: Snap Inc. (Santa Monica, CA)

Family ID: 59581991

Assignee: Snap Inc. (Santa Monica, CA)

Appl. No.: 17/215152

Filed: March 29, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210217217 A1	Jul. 15, 2021

Related U.S. Application Data

continuation parent-doc US 16433725 20190606 US 10984569 child-doc US 17215152
continuation parent-doc US 15199472 20160630 US 10360708 child-doc US 16433725

Publication Classification

Int. Cl.: G06T11/60 (20060101); G06V40/16 (20220101)

U.S. Cl.:

CPC G06T11/60 (20130101); G06V40/171 (20220101); G06V40/172 (20220101); G06V40/176 (20220101); G06T2200/24 (20130101)

Field of Classification Search

CPC: G06T (11/60); G06V (40/176); G06V (40/171); G06V (40/172)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
666223	12/1900	Shedlock	N/A	N/A
4581634	12/1985	Williams	N/A	N/A
4975690	12/1989	Torres	N/A	N/A
5072412	12/1990	Henderson, Jr. et al.	N/A	N/A
5493692	12/1995	Theimer et al.	N/A	N/A
5713073	12/1997	Warsta	N/A	N/A
5754939	12/1997	Herz et al.	N/A	N/A
5826269	12/1997	Hussey	N/A	N/A
5855008	12/1997	Goldhaber et al.	N/A	N/A
5880731	12/1998	Liles et al.	N/A	N/A
5883639	12/1998	Walton et al.	N/A	N/A
5999932	12/1998	Paul	N/A	N/A
6012098	12/1999	Bayeh et al.	N/A	N/A
6014090	12/1999	Rosen et al.	N/A	N/A
6023270	12/1999	Brush et al.	N/A	N/A
6029141	12/1999	Bezos et al.	N/A	N/A
6038295	12/1999	Mattes	N/A	N/A
6049711	12/1999	Yehezkel et al.	N/A	N/A
6154764	12/1999	Nitta et al.	N/A	N/A
6158044	12/1999	Tibbetts	N/A	N/A
6167435	12/1999	Druckenmiller et al.	N/A	N/A
6204840	12/2000	Petelycky et al.	N/A	N/A
6205432	12/2000	Gabbard et al.	N/A	N/A
6216141	12/2000	Straub et al.	N/A	N/A
6223165	12/2000	Lauffer	N/A	N/A
6233318	12/2000	Picard et al.	N/A	N/A
6283858	12/2000	Hayes, Jr. et al.	N/A	N/A
6285381	12/2000	Sawano et al.	N/A	N/A
6285987	12/2000	Roth et al.	N/A	N/A
6310694	12/2000	Okimoto et al.	N/A	N/A
6317789	12/2000	Rakavy et al.	N/A	N/A
6334149	12/2000	Davis, Jr. et al.	N/A	N/A

6349203	12/2001	Asaoka et al.	N/A	N/A
6353170	12/2001	Eyzaguirre et al.	N/A	N/A
6374292	12/2001	Srivastava et al.	N/A	N/A
6446004	12/2001	Cao et al.	N/A	N/A
6449657	12/2001	Stanbach et al.	N/A	N/A
6456852	12/2001	Bar et al.	N/A	N/A
6473794	12/2001	Guheen et al.	N/A	N/A
6484196	12/2001	Maurille	N/A	N/A
6487586	12/2001	Ogilvie et al.	N/A	N/A
6487601	12/2001	Hubacher et al.	N/A	N/A
6523008	12/2002	Avrunin	N/A	N/A
6542749	12/2002	Tanaka et al.	N/A	N/A
6549768	12/2002	Fraccaroli	N/A	N/A
6590674	12/2002	Orton	358/1.18	G06F 16/93
6618593	12/2002	Drutman et al.	N/A	N/A
6622174	12/2002	Ukita et al.	N/A	N/A
6631463	12/2002	Floyd et al.	N/A	N/A
6636247	12/2002	Hamzy et al.	N/A	N/A
6636855	12/2002	Holloway et al.	N/A	N/A
6643684	12/2002	Malkin et al.	N/A	N/A
6658095	12/2002	Yoakum et al.	N/A	N/A
6665531	12/2002	Soderbacka et al.	N/A	N/A
6668173	12/2002	Greene	N/A	N/A
6684238	12/2003	Dutta	N/A	N/A
6684257	12/2003	Camut et al.	N/A	N/A
6698020	12/2003	Zigmond et al.	N/A	N/A
6700506	12/2003	Winkler	N/A	N/A
6701347	12/2003	Ogilvie	N/A	N/A
6711608	12/2003	Ogilvie	N/A	N/A
6720860	12/2003	Narayanaswami	N/A	N/A
6724403	12/2003	Santoro et al.	N/A	N/A
6757713	12/2003	Ogilvie et al.	N/A	N/A
6772195	12/2003	Hatlelid et al.	N/A	N/A
6832222	12/2003	Zimowski	N/A	N/A
6834195	12/2003	Brandenberg et al.	N/A	N/A
6836792	12/2003	Chen	N/A	N/A
6839411	12/2004	Saltanov et al.	N/A	N/A
6842779	12/2004	Nishizawa	N/A	N/A
6898626	12/2004	Ohashi	N/A	N/A
6959324	12/2004	Kubik et al.	N/A	N/A
6970088	12/2004	Kovach	N/A	N/A
6970907	12/2004	Ullmann et al.	N/A	N/A
6980909	12/2004	Root et al.	N/A	N/A
6981040	12/2004	Konig et al.	N/A	N/A
7020494	12/2005	Spiestersbach et al.	N/A	N/A
7027124	12/2005	Foote et al.	N/A	N/A
7072963	12/2005	Anderson et al.	N/A	N/A
7073129	12/2005	Robarts et al.	N/A	N/A
7079158	12/2005	Lambertsen	N/A	N/A
7085571	12/2005	Kalhan et al.	N/A	N/A

7110744	12/2005	Freeny, Jr.	N/A	N/A
7124164	12/2005	Chemtob	N/A	N/A
7149893	12/2005	Leonard et al.	N/A	N/A
7173651	12/2006	Knowles	N/A	N/A
7188143	12/2006	Szeto	N/A	N/A
7203380	12/2006	Chiu et al.	N/A	N/A
7206568	12/2006	Sudit	N/A	N/A
7227937	12/2006	Yoakum et al.	N/A	N/A
7237002	12/2006	Estrada et al.	N/A	N/A
7240089	12/2006	Boudreau	N/A	N/A
7243163	12/2006	Friend et al.	N/A	N/A
7269426	12/2006	Kokkonen et al.	N/A	N/A
7278168	12/2006	Chaudhury et al.	N/A	N/A
7280123	12/2006	Bentley et al.	N/A	N/A
7280658	12/2006	Amini et al.	N/A	N/A
7315823	12/2007	Brondrup	N/A	N/A
7342587	12/2007	Danzig et al.	N/A	N/A
7349768	12/2007	Bruce et al.	N/A	N/A
7356564	12/2007	Hartselle et al.	N/A	N/A
7376715	12/2007	Cunningham et al.	N/A	N/A
7394345	12/2007	Ehlinger et al.	N/A	N/A
7411493	12/2007	Smith	N/A	N/A
7423580	12/2007	Markhovsky et al.	N/A	N/A
7454442	12/2007	Cobleigh et al.	N/A	N/A
7468729	12/2007	Levinson	N/A	N/A
7478402	12/2008	Christensen et al.	N/A	N/A
7496347	12/2008	Puranik	N/A	N/A
7508419	12/2008	Toyama et al.	N/A	N/A
7512649	12/2008	Faybishenko et al.	N/A	N/A
7519670	12/2008	Hagale et al.	N/A	N/A
7535469	12/2008	Kim et al.	N/A	N/A
7535890	12/2008	Rojas	N/A	N/A
7546554	12/2008	Chiu et al.	N/A	N/A
7607096	12/2008	Oreizy et al.	N/A	N/A
7627828	12/2008	Collison et al.	N/A	N/A
7636755	12/2008	Blattner et al.	N/A	N/A
7639251	12/2008	Gu et al.	N/A	N/A
7639943	12/2008	Kalajan	N/A	N/A
7650231	12/2009	Gadler	N/A	N/A
7668537	12/2009	DeVries	N/A	N/A
7689649	12/2009	Heikes et al.	N/A	N/A
7703140	12/2009	Nath et al.	N/A	N/A
7711155	12/2009	Sharma et al.	N/A	N/A
7770137	12/2009	Forbes et al.	N/A	N/A
7775885	12/2009	Van Luchene et al.	N/A	N/A
7778973	12/2009	Choi	N/A	N/A
7779444	12/2009	Glad	N/A	N/A
7787886	12/2009	Markhovsky et al.	N/A	N/A
7792789	12/2009	Prahlad et al.	N/A	N/A
7796946	12/2009	Eisenbach	N/A	N/A

7801954	12/2009	Cadiz et al.	N/A	N/A
7818336	12/2009	Amidon et al.	N/A	N/A
7856360	12/2009	Kramer et al.	N/A	N/A
7859551	12/2009	Bulman et al.	N/A	N/A
7885931	12/2010	Seo et al.	N/A	N/A
7912896	12/2010	Wolovitz et al.	N/A	N/A
7925703	12/2010	Dinan et al.	N/A	N/A
8001204	12/2010	Burtner et al.	N/A	N/A
8032586	12/2010	Challenger et al.	N/A	N/A
8077931	12/2010	Chatman et al.	N/A	N/A
8082255	12/2010	Carlson, Jr. et al.	N/A	N/A
8088044	12/2011	Tchao et al.	N/A	N/A
8090351	12/2011	Klein	N/A	N/A
8095878	12/2011	Bates et al.	N/A	N/A
8098904	12/2011	Ioffe et al.	N/A	N/A
8099109	12/2011	Altman et al.	N/A	N/A
8108774	12/2011	Finn et al.	N/A	N/A
8112716	12/2011	Kobayashi	N/A	N/A
8117281	12/2011	Robinson et al.	N/A	N/A
8130219	12/2011	Fleury et al.	N/A	N/A
8131597	12/2011	Hudetz	N/A	N/A
8135166	12/2011	Rhoads	N/A	N/A
8136028	12/2011	Loeb et al.	N/A	N/A
8146001	12/2011	Reese	N/A	N/A
8146005	12/2011	Jones et al.	N/A	N/A
8151191	12/2011	Nicol	N/A	N/A
8161115	12/2011	Yamamoto	N/A	N/A
8161417	12/2011	Lee	N/A	N/A
8170957	12/2011	Richard	N/A	N/A
8195203	12/2011	Tseng	N/A	N/A
8195748	12/2011	Hallyn	N/A	N/A
8199747	12/2011	Rojas et al.	N/A	N/A
8208943	12/2011	Petersen	N/A	N/A
8214443	12/2011	Hamburg	N/A	N/A
8234350	12/2011	Gu et al.	N/A	N/A
8238947	12/2011	Lottin et al.	N/A	N/A
8244593	12/2011	Klinger et al.	N/A	N/A
8276092	12/2011	Narayanan et al.	N/A	N/A
8279319	12/2011	Date	N/A	N/A
8280406	12/2011	Ziskind et al.	N/A	N/A
8285199	12/2011	Hsu et al.	N/A	N/A
8287380	12/2011	Nguyen et al.	N/A	N/A
8301159	12/2011	Hamynen et al.	N/A	N/A
8306922	12/2011	Kunal et al.	N/A	N/A
8312086	12/2011	Velusamy et al.	N/A	N/A
8312097	12/2011	Siegel et al.	N/A	N/A
8326315	12/2011	Phillips et al.	N/A	N/A
8326327	12/2011	Hymel et al.	N/A	N/A
8332475	12/2011	Rosen et al.	N/A	N/A
8352546	12/2012	Dollard	N/A	N/A

8379130	12/2012	Forutanpour et al.	N/A	N/A
8384719	12/2012	Reville et al.	N/A	N/A
8385950	12/2012	Wagner et al.	N/A	N/A
RE44054	12/2012	Kim	N/A	N/A
8396708	12/2012	Park et al.	N/A	N/A
8402097	12/2012	Szeto	N/A	N/A
8405773	12/2012	Hayashi et al.	N/A	N/A
8413059	12/2012	Lee et al.	N/A	N/A
8418067	12/2012	Cheng et al.	N/A	N/A
8423409	12/2012	Rao	N/A	N/A
8425322	12/2012	Gillo et al.	N/A	N/A
8457367	12/2012	Sipe et al.	N/A	N/A
8458601	12/2012	Castelli et al.	N/A	N/A
8462198	12/2012	Lin et al.	N/A	N/A
8471914	12/2012	Sakiyama et al.	N/A	N/A
8472935	12/2012	Fujisaki	N/A	N/A
8484158	12/2012	Deluca et al.	N/A	N/A
8495503	12/2012	Brown et al.	N/A	N/A
8495505	12/2012	Smith et al.	N/A	N/A
8504926	12/2012	Wolf	N/A	N/A
8510383	12/2012	Hurley et al.	N/A	N/A
8527345	12/2012	Rothschild et al.	N/A	N/A
8554627	12/2012	Svendsen et al.	N/A	N/A
8559980	12/2012	Pujol	N/A	N/A
8560612	12/2012	Kilmer et al.	N/A	N/A
8564621	12/2012	Branson et al.	N/A	N/A
8564710	12/2012	Nonaka et al.	N/A	N/A
8570326	12/2012	Gorev	N/A	N/A
8570907	12/2012	Garcia, Jr. et al.	N/A	N/A
8581911	12/2012	Becker et al.	N/A	N/A
8594680	12/2012	Ledlie et al.	N/A	N/A
8597121	12/2012	Andres del Valle	N/A	N/A
8601051	12/2012	Wang	N/A	N/A
8601379	12/2012	Marks et al.	N/A	N/A
8613089	12/2012	Holloway et al.	N/A	N/A
8632408	12/2013	Gillo et al.	N/A	N/A
8648865	12/2013	Dawson et al.	N/A	N/A
8655389	12/2013	Jackson et al.	N/A	N/A
8659548	12/2013	Hildreth	N/A	N/A
8660358	12/2013	Bergboer et al.	N/A	N/A
8660369	12/2013	Llano et al.	N/A	N/A
8660793	12/2013	Ngo et al.	N/A	N/A
8682350	12/2013	Altman et al.	N/A	N/A
8683354	12/2013	Khandelwal et al.	N/A	N/A
8692830	12/2013	Nelson et al.	N/A	N/A
8700012	12/2013	Ferren et al.	N/A	N/A
8718333	12/2013	Wolf et al.	N/A	N/A
8724622	12/2013	Rojas	N/A	N/A
8730231	12/2013	Snoddy et al.	N/A	N/A
8732168	12/2013	Johnson	N/A	N/A

8738719	12/2013	Lee et al.	N/A	N/A
8744523	12/2013	Fan et al.	N/A	N/A
8745132	12/2013	Obradovich	N/A	N/A
8761800	12/2013	Kuwahara	N/A	N/A
8768876	12/2013	Shim et al.	N/A	N/A
8775972	12/2013	Spiegel	N/A	N/A
8788680	12/2013	Naik	N/A	N/A
8790187	12/2013	Walker et al.	N/A	N/A
8797415	12/2013	Arnold	N/A	N/A
8798646	12/2013	Wang et al.	N/A	N/A
8810513	12/2013	Ptucha et al.	N/A	N/A
8812171	12/2013	Filev et al.	N/A	N/A
8832201	12/2013	Wall	N/A	N/A
8832552	12/2013	Arrasvuori et al.	N/A	N/A
8839327	12/2013	Amento et al.	N/A	N/A
8856349	12/2013	Jain et al.	N/A	N/A
8874677	12/2013	Rosen et al.	N/A	N/A
8886227	12/2013	Schmidt et al.	N/A	N/A
8887035	12/2013	Mcdonald et al.	N/A	N/A
8890926	12/2013	Tandon et al.	N/A	N/A
8892999	12/2013	Nims et al.	N/A	N/A
8893010	12/2013	Brin et al.	N/A	N/A
8909679	12/2013	Root et al.	N/A	N/A
8909714	12/2013	Agarwal et al.	N/A	N/A
8909725	12/2013	Sehn	N/A	N/A
8914752	12/2013	Spiegel	N/A	N/A
8924250	12/2013	Bates et al.	N/A	N/A
8935656	12/2014	Dandia et al.	N/A	N/A
8963926	12/2014	Brown et al.	N/A	N/A
8972357	12/2014	Shim et al.	N/A	N/A
8989786	12/2014	Feghali	N/A	N/A
8995433	12/2014	Rojas	N/A	N/A
9002643	12/2014	Xu	N/A	N/A
9015285	12/2014	Ebsen et al.	N/A	N/A
9020745	12/2014	Johnston et al.	N/A	N/A
9040574	12/2014	Wang et al.	N/A	N/A
9055416	12/2014	Rosen et al.	N/A	N/A
9083770	12/2014	Drose et al.	N/A	N/A
9086776	12/2014	Ye et al.	N/A	N/A
9094137	12/2014	Sehn et al.	N/A	N/A
9100806	12/2014	Rosen et al.	N/A	N/A
9100807	12/2014	Rosen et al.	N/A	N/A
9105014	12/2014	Collet et al.	N/A	N/A
9113301	12/2014	Spiegel et al.	N/A	N/A
9119027	12/2014	Sharon et al.	N/A	N/A
9123074	12/2014	Jacobs et al.	N/A	N/A
9135726	12/2014	Kafuku	N/A	N/A
9143382	12/2014	Bhogal et al.	N/A	N/A
9143681	12/2014	Ebsen et al.	N/A	N/A
9148424	12/2014	Yang	N/A	N/A

9152477	12/2014	Campbell et al.	N/A	N/A
9191776	12/2014	Root et al.	N/A	N/A
9204252	12/2014	Root	N/A	N/A
9224220	12/2014	Toyoda et al.	N/A	N/A
9225805	12/2014	Kujawa et al.	N/A	N/A
9225897	12/2014	Sehn et al.	N/A	N/A
9237202	12/2015	Sehn	N/A	N/A
9241184	12/2015	Weerasinghe	N/A	N/A
9256860	12/2015	Herger et al.	N/A	N/A
9256950	12/2015	Xu	N/A	G06V 40/171
9258459	12/2015	Hartley	N/A	N/A
9264463	12/2015	Rubinstein et al.	N/A	N/A
9276886	12/2015	Samaranayake	N/A	N/A
9285951	12/2015	Makofsky et al.	N/A	N/A
9294425	12/2015	Son	N/A	N/A
9298257	12/2015	Hwang et al.	N/A	N/A
9314692	12/2015	Konoplev et al.	N/A	N/A
9330483	12/2015	Du et al.	N/A	N/A
9344606	12/2015	Hartley et al.	N/A	N/A
9357174	12/2015	Li et al.	N/A	N/A
9361510	12/2015	Yao et al.	N/A	N/A
9378576	12/2015	Bouaziz et al.	N/A	N/A
9380325	12/2015	Cormie et al.	N/A	N/A
9385983	12/2015	Sehn	N/A	N/A
9392308	12/2015	Ahmed et al.	N/A	N/A
9396354	12/2015	Murphy et al.	N/A	N/A
9402057	12/2015	Kaytaz et al.	N/A	N/A
9407712	12/2015	Sehn	N/A	N/A
9407816	12/2015	Sehn	N/A	N/A
9412192	12/2015	Mandel et al.	N/A	N/A
9430783	12/2015	Sehn	N/A	N/A
9439041	12/2015	Parvizi et al.	N/A	N/A
9443227	12/2015	Evans et al.	N/A	N/A
9450907	12/2015	Pridmore et al.	N/A	N/A
9459778	12/2015	Hogeg et al.	N/A	N/A
9460541	12/2015	Li et al.	N/A	N/A
9480924	12/2015	Haslam	N/A	N/A
9482882	12/2015	Hanover et al.	N/A	N/A
9482883	12/2015	Meisenholder	N/A	N/A
9485747	12/2015	Rodoper et al.	N/A	N/A
9489661	12/2015	Evans et al.	N/A	N/A
9489760	12/2015	Li et al.	N/A	N/A
9491134	12/2015	Rosen et al.	N/A	N/A
9503845	12/2015	Vincent	N/A	N/A
9508197	12/2015	Quinn et al.	N/A	N/A
9532171	12/2015	Allen et al.	N/A	N/A
9537811	12/2016	Allen et al.	N/A	N/A
9544257	12/2016	Ogundokun et al.	N/A	N/A
9560006	12/2016	Prado et al.	N/A	N/A

9576400	12/2016	Van Os et al.	N/A	N/A
9589357	12/2016	Li et al.	N/A	N/A
9592449	12/2016	Barbalet et al.	N/A	N/A
9628950	12/2016	Noeth et al.	N/A	N/A
9635195	12/2016	Green et al.	N/A	N/A
9641870	12/2016	Cormie et al.	N/A	N/A
9648376	12/2016	Chang et al.	N/A	N/A
9652896	12/2016	Jurgenson et al.	N/A	N/A
9659244	12/2016	Anderton et al.	N/A	N/A
9693191	12/2016	Sehn	N/A	N/A
9697635	12/2016	Quinn et al.	N/A	N/A
9705831	12/2016	Spiegel	N/A	N/A
9706040	12/2016	Kadirvel et al.	N/A	N/A
9710821	12/2016	Heath	N/A	N/A
9742713	12/2016	Spiegel et al.	N/A	N/A
9744466	12/2016	Fujioka	N/A	N/A
9746990	12/2016	Anderson et al.	N/A	N/A
9749270	12/2016	Collet et al.	N/A	N/A
9773284	12/2016	Huang et al.	N/A	N/A
9785796	12/2016	Murphy et al.	N/A	N/A
9792714	12/2016	Li et al.	N/A	N/A
9824463	12/2016	Ingrassia et al.	N/A	N/A
9825898	12/2016	Sehn	N/A	N/A
9839844	12/2016	Dunstan et al.	N/A	N/A
9854219	12/2016	Sehn	N/A	N/A
9883838	12/2017	Kaleal, III et al.	N/A	N/A
9898849	12/2017	Du et al.	N/A	N/A
9911073	12/2017	Spiegel et al.	N/A	N/A
9936165	12/2017	Li et al.	N/A	N/A
9959037	12/2017	Chaudhri et al.	N/A	N/A
9961520	12/2017	Brooks et al.	N/A	N/A
9980100	12/2017	Charlton et al.	N/A	N/A
9990373	12/2017	Fortkort	N/A	N/A
10039988	12/2017	Lobb et al.	N/A	N/A
10097492	12/2017	Tsuda et al.	N/A	N/A
10116598	12/2017	Tucker et al.	N/A	N/A
10121055	12/2017	Savvides et al.	N/A	N/A
10127945	12/2017	Ho et al.	N/A	N/A
10146748	12/2017	Barndollar et al.	N/A	N/A
10155168	12/2017	Blackstock et al.	N/A	N/A
10158589	12/2017	Collet et al.	N/A	N/A
10178507	12/2018	Roberts	N/A	N/A
10194270	12/2018	Yokoyama et al.	N/A	N/A
10212541	12/2018	Brody et al.	N/A	N/A
10237692	12/2018	Shan et al.	N/A	N/A
10242477	12/2018	Charlton et al.	N/A	N/A
10242503	12/2018	McPhee et al.	N/A	N/A
10262250	12/2018	Spiegel et al.	N/A	N/A
10339365	12/2018	Gusarov et al.	N/A	N/A
10360708	12/2018	Bondich et al.	N/A	N/A

10362219	12/2018	Wilson et al.	N/A	N/A
10375519	12/2018	Pai et al.	N/A	N/A
10382378	12/2018	Garcia et al.	N/A	N/A
10432498	12/2018	Mcclendon	N/A	N/A
10432559	12/2018	Baldwin et al.	N/A	N/A
10454857	12/2018	Blackstock et al.	N/A	N/A
10475225	12/2018	Park et al.	N/A	N/A
10496661	12/2018	Morgan et al.	N/A	N/A
10504266	12/2018	Blattner et al.	N/A	N/A
10573048	12/2019	Ni et al.	N/A	N/A
10657701	12/2019	Osman et al.	N/A	N/A
10803527	12/2019	Zankat et al.	N/A	N/A
10880246	12/2019	Baldwin et al.	N/A	N/A
10938758	12/2020	Allen et al.	N/A	N/A
10952013	12/2020	Brody et al.	N/A	N/A
10963529	12/2020	Amitay et al.	N/A	N/A
10984569	12/2020	Bondich et al.	N/A	N/A
11843456	12/2022	Allen et al.	N/A	N/A
11876762	12/2023	Allen et al.	N/A	N/A
12113760	12/2023	Allen et al.	N/A	N/A
2002/0035607	12/2001	Checkoway et al.	N/A	N/A
2002/0047868	12/2001	Miyazawa	N/A	N/A
2002/0059193	12/2001	Decime et al.	N/A	N/A
2002/0067362	12/2001	Agostino Nocera et al.	N/A	N/A
2002/0078456	12/2001	Hudson et al.	N/A	N/A
2002/0087631	12/2001	Sharma	N/A	N/A
2002/0097257	12/2001	Miller et al.	N/A	N/A
2002/0122659	12/2001	Mcgrath et al.	N/A	N/A
2002/0128047	12/2001	Gates	N/A	N/A
2002/0144154	12/2001	Tomkow	N/A	N/A
2002/0169644	12/2001	Greene	N/A	N/A
2003/0001846	12/2002	Davis et al.	N/A	N/A
2003/0016247	12/2002	Lai et al.	N/A	N/A
2003/0017823	12/2002	Mager et al.	N/A	N/A
2003/0020623	12/2002	Cao et al.	N/A	N/A
2003/0023874	12/2002	Prokupets et al.	N/A	N/A
2003/0037124	12/2002	Yamaura et al.	N/A	N/A
2003/0052925	12/2002	Daimon et al.	N/A	N/A
2003/0101230	12/2002	Benschoter et al.	N/A	N/A
2003/0110503	12/2002	Perkes	N/A	N/A
2003/0126215	12/2002	Udell	N/A	N/A
2003/0148773	12/2002	Spriestersbach et al.	N/A	N/A
2003/0164856	12/2002	Prager et al.	N/A	N/A
2003/0206171	12/2002	Kim et al.	N/A	N/A
2003/0217106	12/2002	Adar et al.	N/A	N/A
2003/0229607	12/2002	Zellweger et al.	N/A	N/A
2004/0027371	12/2003	Jaeger	N/A	N/A
2004/0064429	12/2003	Hirstius et al.	N/A	N/A
2004/0078367	12/2003	Anderson et al.	N/A	N/A

2004/0111467	12/2003	Willis	N/A	N/A
2004/0158739	12/2003	Wakai et al.	N/A	N/A
2004/0189465	12/2003	Capobianco et al.	N/A	N/A
2004/0203959	12/2003	Coombes	N/A	N/A
2004/0215625	12/2003	Svendsen et al.	N/A	N/A
2004/0243531	12/2003	Dean	N/A	N/A
2004/0243688	12/2003	Wugofski	N/A	N/A
2005/0021444	12/2004	Bauer et al.	N/A	N/A
2005/0022211	12/2004	Veselov et al.	N/A	N/A
2005/0048989	12/2004	Jung	N/A	N/A
2005/0078804	12/2004	Yomoda	N/A	N/A
2005/0097176	12/2004	Schatz et al.	N/A	N/A
2005/0102381	12/2004	Jiang et al.	N/A	N/A
2005/0104976	12/2004	Currans	N/A	N/A
2005/0114783	12/2004	Szeto	N/A	N/A
2005/0119936	12/2004	Buchanan et al.	N/A	N/A
2005/0122405	12/2004	Voss et al.	N/A	N/A
2005/0143136	12/2004	Lev et al.	N/A	N/A
2005/0144241	12/2004	Stata et al.	N/A	N/A
2005/0162419	12/2004	Kim et al.	N/A	N/A
2005/0193340	12/2004	Amburgey et al.	N/A	N/A
2005/0193345	12/2004	Klassen et al.	N/A	N/A
2005/0198128	12/2004	Anderson	N/A	N/A
2005/0206610	12/2004	Cordelli	N/A	N/A
2005/0223066	12/2004	Buchheit et al.	N/A	N/A
2005/0280660	12/2004	Seo et al.	N/A	N/A
2005/0288954	12/2004	McCarthy et al.	N/A	N/A
2006/0026067	12/2005	Nicholas et al.	N/A	N/A
2006/0031412	12/2005	Adams et al.	N/A	N/A
2006/0107297	12/2005	Toyama et al.	N/A	N/A
2006/0114338	12/2005	Rothschild	N/A	N/A
2006/0119882	12/2005	Harris et al.	N/A	N/A
2006/0145944	12/2005	Tarlton et al.	N/A	N/A
2006/0242239	12/2005	Morishima et al.	N/A	N/A
2006/0252438	12/2005	Ansamaa et al.	N/A	N/A
2006/0265417	12/2005	Amato et al.	N/A	N/A
2006/0270419	12/2005	Crowley et al.	N/A	N/A
2006/0287878	12/2005	Wadhwa et al.	N/A	N/A
2006/0294465	12/2005	Ronen et al.	N/A	N/A
2007/0004426	12/2006	Pfleging et al.	N/A	N/A
2007/0011270	12/2006	Klein et al.	N/A	N/A
2007/0038715	12/2006	Collins et al.	N/A	N/A
2007/0040931	12/2006	Nishizawa	N/A	N/A
2007/0064899	12/2006	Boss et al.	N/A	N/A
2007/0073517	12/2006	Panje	N/A	N/A
2007/0073823	12/2006	Cohen et al.	N/A	N/A
2007/0075898	12/2006	Markhovsky et al.	N/A	N/A
2007/0082707	12/2006	Flynt et al.	N/A	N/A
2007/0113181	12/2006	Blattner et al.	N/A	N/A
2007/0136228	12/2006	Petersen	N/A	N/A

2007/0168863	12/2006	Blattner et al.	N/A	N/A
2007/0174273	12/2006	Jones et al.	N/A	N/A
2007/0176921	12/2006	Iwasaki et al.	N/A	N/A
2007/0192128	12/2006	Celestini	N/A	N/A
2007/0198340	12/2006	Lucovsky et al.	N/A	N/A
2007/0198495	12/2006	Buron et al.	N/A	N/A
2007/0208751	12/2006	Cowan et al.	N/A	N/A
2007/0210936	12/2006	Nicholson	N/A	N/A
2007/0214180	12/2006	Crawford	N/A	N/A
2007/0214216	12/2006	Carrer et al.	N/A	N/A
2007/0218987	12/2006	Luchene et al.	N/A	N/A
2007/0233556	12/2006	Koningstein	N/A	N/A
2007/0233801	12/2006	Eren et al.	N/A	N/A
2007/0233859	12/2006	Zhao et al.	N/A	N/A
2007/0243887	12/2006	Bandhole et al.	N/A	N/A
2007/0244750	12/2006	Grannan et al.	N/A	N/A
2007/0255456	12/2006	Funayama	N/A	N/A
2007/0258656	12/2006	Aarabi et al.	N/A	N/A
2007/0260984	12/2006	Marks et al.	N/A	N/A
2007/0281690	12/2006	Altman et al.	N/A	N/A
2008/0022329	12/2007	Glad	N/A	N/A
2008/0025701	12/2007	Ikeda	N/A	N/A
2008/0032703	12/2007	Krumm et al.	N/A	N/A
2008/0033930	12/2007	Warren	N/A	N/A
2008/0043041	12/2007	Hedenstroem et al.	N/A	N/A
2008/0049704	12/2007	Witteman et al.	N/A	N/A
2008/0055269	12/2007	Lemay et al.	N/A	N/A
2008/0062141	12/2007	Chandhri	N/A	N/A
2008/0070593	12/2007	Altman et al.	N/A	N/A
2008/0076505	12/2007	Nguyen et al.	N/A	N/A
2008/0092233	12/2007	Tian et al.	N/A	N/A
2008/0094387	12/2007	Chen	N/A	N/A
2008/0097979	12/2007	Heidloff et al.	N/A	N/A
2008/0104503	12/2007	Beall et al.	N/A	N/A
2008/0109844	12/2007	Baldeschweiler et al.	N/A	N/A
2008/0120409	12/2007	Sun et al.	N/A	N/A
2008/0147730	12/2007	Lee et al.	N/A	N/A
2008/0148150	12/2007	Mall	N/A	N/A
2008/0158222	12/2007	Li et al.	N/A	N/A
2008/0158230	12/2007	Sharma et al.	N/A	N/A
2008/0168033	12/2007	Ott et al.	N/A	N/A
2008/0168489	12/2007	Schraga	N/A	N/A
2008/0189177	12/2007	Anderton et al.	N/A	N/A
2008/0201638	12/2007	Nair	N/A	N/A
2008/0207176	12/2007	Brackbill et al.	N/A	N/A
2008/0208692	12/2007	Garaventi et al.	N/A	N/A
2008/0209329	12/2007	Defranco et al.	N/A	N/A
2008/0214210	12/2007	Rasanen et al.	N/A	N/A
2008/0216092	12/2007	Serlet	N/A	N/A
2008/0222108	12/2007	Prahlad et al.	N/A	N/A

2008/0222545	12/2007	Lemay	N/A	N/A
2008/0255976	12/2007	Altberg et al.	N/A	N/A
2008/0256446	12/2007	Yamamoto	N/A	N/A
2008/0256577	12/2007	Funaki et al.	N/A	N/A
2008/0266421	12/2007	Takahata et al.	N/A	N/A
2008/0270938	12/2007	Carlson	N/A	N/A
2008/0288338	12/2007	Wiseman et al.	N/A	N/A
2008/0306826	12/2007	Kramer et al.	N/A	N/A
2008/0309617	12/2007	Kong et al.	N/A	N/A
2008/0309675	12/2007	Fleury	345/581	G06T 17/00
2008/0313329	12/2007	Wang et al.	N/A	N/A
2008/0313346	12/2007	Kujawa et al.	N/A	N/A
2008/0318616	12/2007	Chipalkatti et al.	N/A	N/A
2009/0006191	12/2008	Arankalle et al.	N/A	N/A
2009/0006565	12/2008	Velusamy et al.	N/A	N/A
2009/0013268	12/2008	Amit	N/A	N/A
2009/0015703	12/2008	Kim et al.	N/A	N/A
2009/0016617	12/2008	Bregman-Amitai et al.	N/A	N/A
2009/0024956	12/2008	Kobayashi	N/A	N/A
2009/0030774	12/2008	Rothschild et al.	N/A	N/A
2009/0030884	12/2008	Pulfer et al.	N/A	N/A
2009/0030999	12/2008	Gatzke et al.	N/A	N/A
2009/0040324	12/2008	Nonaka	N/A	N/A
2009/0042588	12/2008	Lottin et al.	N/A	N/A
2009/0044113	12/2008	Jones et al.	N/A	N/A
2009/0047972	12/2008	Neeraj	N/A	N/A
2009/0055484	12/2008	Vuong et al.	N/A	N/A
2009/0058822	12/2008	Chaudhri	N/A	N/A
2009/0070688	12/2008	Gyorfi et al.	N/A	N/A
2009/0079743	12/2008	Pearson et al.	N/A	N/A
2009/0079846	12/2008	Chou	N/A	N/A
2009/0087035	12/2008	Wen et al.	N/A	N/A
2009/0089678	12/2008	Sacco et al.	N/A	N/A
2009/0089710	12/2008	Wood et al.	N/A	N/A
2009/0093261	12/2008	Ziskind	N/A	N/A
2009/0099925	12/2008	Mehta et al.	N/A	N/A
2009/0100367	12/2008	Dargahi et al.	N/A	N/A
2009/0106672	12/2008	Burstrom	N/A	N/A
2009/0132341	12/2008	Klinger	N/A	N/A
2009/0132453	12/2008	Hangartner et al.	N/A	N/A
2009/0132665	12/2008	Thomsen et al.	N/A	N/A
2009/0144639	12/2008	Nims et al.	N/A	N/A
2009/0148045	12/2008	Lee et al.	N/A	N/A
2009/0150778	12/2008	Nicol	N/A	N/A
2009/0153492	12/2008	Popp	N/A	N/A
2009/0153552	12/2008	Fidaleo	345/419	G06Q 30/02
2009/0157450	12/2008	Athsani et al.	N/A	N/A
2009/0157752	12/2008	Gonzalez	N/A	N/A
2009/0158170	12/2008	Narayanan et al.	N/A	N/A

2009/0160970	12/2008	Fredlund et al.	N/A	N/A
2009/0163182	12/2008	Gatti et al.	N/A	N/A
2009/0175521	12/2008	Shadan et al.	N/A	N/A
2009/0177299	12/2008	Van De Sluis	N/A	N/A
2009/0177976	12/2008	Bokor et al.	N/A	N/A
2009/0192900	12/2008	Collision	N/A	N/A
2009/0199242	12/2008	Johnson et al.	N/A	N/A
2009/0202114	12/2008	Morin et al.	N/A	N/A
2009/0215469	12/2008	Fisher et al.	N/A	N/A
2009/0228811	12/2008	Adams et al.	N/A	N/A
2009/0232354	12/2008	Camp, Jr. et al.	N/A	N/A
2009/0234815	12/2008	Boerries et al.	N/A	N/A
2009/0239552	12/2008	Churchill et al.	N/A	N/A
2009/0249222	12/2008	Schmidt et al.	N/A	N/A
2009/0249244	12/2008	Robinson et al.	N/A	N/A
2009/0254840	12/2008	Churchill et al.	N/A	N/A
2009/0254859	12/2008	Arrasvuori et al.	N/A	N/A
2009/0265604	12/2008	Howard et al.	N/A	N/A
2009/0265647	12/2008	Martin et al.	N/A	N/A
2009/0284551	12/2008	Stanton	N/A	N/A
2009/0288022	12/2008	Almstrand et al.	N/A	N/A
2009/0291672	12/2008	Treves et al.	N/A	N/A
2009/0292608	12/2008	Polachek	N/A	N/A
2009/0300525	12/2008	Jolliff et al.	N/A	N/A
2009/0303984	12/2008	Clark et al.	N/A	N/A
2009/0319178	12/2008	Khosravy et al.	N/A	N/A
2009/0319607	12/2008	Belz et al.	N/A	N/A
2009/0327073	12/2008	Li	N/A	N/A
2009/0328122	12/2008	Amento et al.	N/A	N/A
2010/0011422	12/2009	Mason et al.	N/A	N/A
2010/0023885	12/2009	Reville et al.	N/A	N/A
2010/0026698	12/2009	Reville	345/581	A63F 13/10
2010/0058212	12/2009	Belitz et al.	N/A	N/A
2010/0062794	12/2009	Han	N/A	N/A
2010/0073458	12/2009	Pace	N/A	N/A
2010/0082427	12/2009	Burgener et al.	N/A	N/A
2010/0082693	12/2009	Hugg et al.	N/A	N/A
2010/0083138	12/2009	Dawson et al.	N/A	N/A
2010/0083140	12/2009	Dawson et al.	N/A	N/A
2010/0083148	12/2009	Finn et al.	N/A	N/A
2010/0100568	12/2009	Papin et al.	N/A	N/A
2010/0100828	12/2009	Khandelwal et al.	N/A	N/A
2010/0113065	12/2009	Narayan et al.	N/A	N/A
2010/0115426	12/2009	Liu et al.	N/A	N/A
2010/0121915	12/2009	Wang	N/A	N/A
2010/0130233	12/2009	Parker	N/A	N/A
2010/0131880	12/2009	Lee et al.	N/A	N/A
2010/0131895	12/2009	Wohlert	N/A	N/A
2010/0146407	12/2009	Bokor et al.	N/A	N/A
2010/0153144	12/2009	Miller et al.	N/A	N/A

2010/0153868	12/2009	Allen et al.	N/A	N/A
2010/0159944	12/2009	Pascal et al.	N/A	N/A
2010/0161658	12/2009	Hamynen et al.	N/A	N/A
2010/0161831	12/2009	Haas et al.	N/A	N/A
2010/0162149	12/2009	Sheleheda et al.	N/A	N/A
2010/0179953	12/2009	Kan et al.	N/A	N/A
2010/0179991	12/2009	Lorch et al.	N/A	N/A
2010/0183280	12/2009	Beauregard et al.	N/A	N/A
2010/0185552	12/2009	Deluca et al.	N/A	N/A
2010/0185640	12/2009	Dettinger et al.	N/A	N/A
2010/0185665	12/2009	Horn et al.	N/A	N/A
2010/0191631	12/2009	Weidmann	N/A	N/A
2010/0197318	12/2009	Petersen et al.	N/A	N/A
2010/0197319	12/2009	Petersen et al.	N/A	N/A
2010/0197396	12/2009	Fujii et al.	N/A	N/A
2010/0198683	12/2009	Aarabi	N/A	N/A
2010/0198694	12/2009	Muthukrishnan	N/A	N/A
2010/0198826	12/2009	Petersen et al.	N/A	N/A
2010/0198828	12/2009	Petersen et al.	N/A	N/A
2010/0198862	12/2009	Jennings et al.	N/A	N/A
2010/0198870	12/2009	Petersen et al.	N/A	N/A
2010/0198917	12/2009	Petersen et al.	N/A	N/A
2010/0201482	12/2009	Robertson et al.	N/A	N/A
2010/0201536	12/2009	Robertson et al.	N/A	N/A
2010/0203968	12/2009	Gill et al.	N/A	N/A
2010/0214436	12/2009	Kim et al.	N/A	N/A
2010/0223128	12/2009	Dukellis et al.	N/A	N/A
2010/0223343	12/2009	Bosan et al.	N/A	N/A
2010/0227682	12/2009	Reville et al.	N/A	N/A
2010/0250109	12/2009	Johnston et al.	N/A	N/A
2010/0257196	12/2009	Waters et al.	N/A	N/A
2010/0259386	12/2009	Holley et al.	N/A	N/A
2010/0262915	12/2009	Bocking et al.	N/A	N/A
2010/0273509	12/2009	Sweeney et al.	N/A	N/A
2010/0274724	12/2009	Bible, Jr. et al.	N/A	N/A
2010/0279713	12/2009	Dicke	N/A	N/A
2010/0281045	12/2009	Dean	N/A	N/A
2010/0290756	12/2009	Karaoguz et al.	N/A	N/A
2010/0299060	12/2009	Snavely et al.	N/A	N/A
2010/0306669	12/2009	Della Pasqua	N/A	N/A
2010/0332980	12/2009	Sun et al.	N/A	N/A
2011/0004071	12/2010	Faiola et al.	N/A	N/A
2011/0010205	12/2010	Richards	N/A	N/A
2011/0022965	12/2010	Lawrence et al.	N/A	N/A
2011/0029512	12/2010	Folgnier et al.	N/A	N/A
2011/0040783	12/2010	Uemichi et al.	N/A	N/A
2011/0040804	12/2010	Peirce et al.	N/A	N/A
2011/0047404	12/2010	Metzler et al.	N/A	N/A
2011/0047505	12/2010	Fillion et al.	N/A	N/A
2011/0050909	12/2010	Ellenby et al.	N/A	N/A

2011/0050915	12/2010	Wang et al.	N/A	N/A
2011/0064388	12/2010	Brown et al.	N/A	N/A
2011/0066664	12/2010	Goldman et al.	N/A	N/A
2011/0066743	12/2010	Hurley et al.	N/A	N/A
2011/0083101	12/2010	Sharon et al.	N/A	N/A
2011/0093780	12/2010	Dunn	N/A	N/A
2011/0099507	12/2010	Nesladek et al.	N/A	N/A
2011/0102630	12/2010	Rukes	N/A	N/A
2011/0113323	12/2010	Fillion et al.	N/A	N/A
2011/0115798	12/2010	Nayar et al.	N/A	N/A
2011/0119133	12/2010	Igelman et al.	N/A	N/A
2011/0126096	12/2010	Ohashi et al.	N/A	N/A
2011/0137881	12/2010	Cheng et al.	N/A	N/A
2011/0145564	12/2010	Moshir et al.	N/A	N/A
2011/0148864	12/2010	Lee et al.	N/A	N/A
2011/0153759	12/2010	Rathod	N/A	N/A
2011/0159890	12/2010	Fortescue et al.	N/A	N/A
2011/0161076	12/2010	Davis et al.	N/A	N/A
2011/0164163	12/2010	Bilbrey et al.	N/A	N/A
2011/0167125	12/2010	Achlioptas	N/A	N/A
2011/0197194	12/2010	D'Angelo et al.	N/A	N/A
2011/0202598	12/2010	Evans et al.	N/A	N/A
2011/0202968	12/2010	Nurmi	N/A	N/A
2011/0211534	12/2010	Schmidt et al.	N/A	N/A
2011/0211764	12/2010	Krupka et al.	N/A	N/A
2011/0213845	12/2010	Logan et al.	N/A	N/A
2011/0215966	12/2010	Kim et al.	N/A	N/A
2011/0225048	12/2010	Nair	N/A	N/A
2011/0238762	12/2010	Soni et al.	N/A	N/A
2011/0238763	12/2010	Shin et al.	N/A	N/A
2011/0239136	12/2010	Goldman et al.	N/A	N/A
2011/0239143	12/2010	Ye et al.	N/A	N/A
2011/0246330	12/2010	Tikku et al.	N/A	N/A
2011/0248992	12/2010	Van Os et al.	N/A	N/A
2011/0249891	12/2010	Li	N/A	N/A
2011/0255736	12/2010	Thompson et al.	N/A	N/A
2011/0273575	12/2010	Lee	N/A	N/A
2011/0282799	12/2010	Huston	N/A	N/A
2011/0283188	12/2010	Farrenkopf	N/A	N/A
2011/0285703	12/2010	Jin	N/A	N/A
2011/0286586	12/2010	Saylor et al.	N/A	N/A
2011/0292051	12/2010	Nelson et al.	N/A	N/A
2011/0300837	12/2010	Misiag	N/A	N/A
2011/0314419	12/2010	Dunn et al.	N/A	N/A
2011/0320373	12/2010	Lee et al.	N/A	N/A
2012/0013770	12/2011	Stafford et al.	N/A	N/A
2012/0015673	12/2011	Klassen et al.	N/A	N/A
2012/0028659	12/2011	Whitney et al.	N/A	N/A
2012/0033718	12/2011	Kauffman et al.	N/A	N/A
2012/0036015	12/2011	Sheikh	N/A	N/A

2012/0036443	12/2011	Ohmori et al.	N/A	N/A
2012/0054797	12/2011	Skog et al.	N/A	N/A
2012/0059722	12/2011	Rao	N/A	N/A
2012/0062805	12/2011	Candelore	N/A	N/A
2012/0069028	12/2011	Bouguerra	N/A	N/A
2012/0084731	12/2011	Filman et al.	N/A	N/A
2012/0084835	12/2011	Thomas et al.	N/A	N/A
2012/0099800	12/2011	Llano et al.	N/A	N/A
2012/0108293	12/2011	Law et al.	N/A	N/A
2012/0110002	12/2011	Giambalvo	707/769	G06F 16/2455
2012/0110096	12/2011	Smarr et al.	N/A	N/A
2012/0113106	12/2011	Choi et al.	N/A	N/A
2012/0113143	12/2011	Adhikari et al.	N/A	N/A
2012/0113272	12/2011	Hata	N/A	N/A
2012/0123830	12/2011	Svendsen et al.	N/A	N/A
2012/0123871	12/2011	Svendsen et al.	N/A	N/A
2012/0123875	12/2011	Svendsen et al.	N/A	N/A
2012/0124126	12/2011	Alcazar et al.	N/A	N/A
2012/0124176	12/2011	Curtis et al.	N/A	N/A
2012/0124458	12/2011	Cruzada	N/A	N/A
2012/0130717	12/2011	Xu et al.	N/A	N/A
2012/0131507	12/2011	Sparandara et al.	N/A	N/A
2012/0131512	12/2011	Takeuchi et al.	N/A	N/A
2012/0139830	12/2011	Hwang et al.	N/A	N/A
2012/0141046	12/2011	Chen et al.	N/A	N/A
2012/0143760	12/2011	Abulafia et al.	N/A	N/A
2012/0150978	12/2011	Monaco	N/A	N/A
2012/0154633	12/2011	Rodriguez	707/769	H04M 1/72454
2012/0165100	12/2011	Lalancette et al.	N/A	N/A
2012/0166971	12/2011	Sachson et al.	N/A	N/A
2012/0169855	12/2011	Oh	N/A	N/A
2012/0172062	12/2011	Altman et al.	N/A	N/A
2012/0173991	12/2011	Roberts et al.	N/A	N/A
2012/0176401	12/2011	Hayward et al.	N/A	N/A
2012/0184248	12/2011	Speede	N/A	N/A
2012/0197724	12/2011	Kendall	N/A	N/A
2012/0200743	12/2011	Blanchflower et al.	N/A	N/A
2012/0209921	12/2011	Adafin et al.	N/A	N/A
2012/0209924	12/2011	Evans et al.	N/A	N/A
2012/0210244	12/2011	De Francisco Lopez et al.	N/A	N/A
2012/0212632	12/2011	Mate et al.	N/A	N/A
2012/0215879	12/2011	Bozo	N/A	N/A
2012/0220264	12/2011	Kawabata	N/A	N/A
2012/0223940	12/2011	Dunstan et al.	N/A	N/A
2012/0226748	12/2011	Bosworth et al.	N/A	N/A
2012/0229506	12/2011	Nishikawa	N/A	N/A
2012/0233000	12/2011	Fisher et al.	N/A	N/A

2012/0236162	12/2011	Imamura	N/A	N/A
2012/0239761	12/2011	Linner et al.	N/A	N/A
2012/0250951	12/2011	Chen	N/A	N/A
2012/0252418	12/2011	Kandekar et al.	N/A	N/A
2012/0254325	12/2011	Majeti et al.	N/A	N/A
2012/0271883	12/2011	Montoya et al.	N/A	N/A
2012/0278387	12/2011	Garcia et al.	N/A	N/A
2012/0278692	12/2011	Shi	N/A	N/A
2012/0290637	12/2011	Perantatos et al.	N/A	N/A
2012/0290977	12/2011	Devecka	N/A	N/A
2012/0299954	12/2011	Wada et al.	N/A	N/A
2012/0304052	12/2011	Tanaka et al.	N/A	N/A
2012/0304080	12/2011	Wormald et al.	N/A	N/A
2012/0307096	12/2011	Ford et al.	N/A	N/A
2012/0307112	12/2011	Kunishige et al.	N/A	N/A
2012/0309520	12/2011	Evertt et al.	N/A	N/A
2012/0315987	12/2011	Walling	N/A	N/A
2012/0319904	12/2011	Lee et al.	N/A	N/A
2012/0323933	12/2011	He et al.	N/A	N/A
2012/0324018	12/2011	Metcalf et al.	N/A	N/A
2013/0006759	12/2012	Srivastava et al.	N/A	N/A
2013/0024757	12/2012	Doll et al.	N/A	N/A
2013/0031180	12/2012	Abendroth et al.	N/A	N/A
2013/0036165	12/2012	Tseng et al.	N/A	N/A
2013/0036364	12/2012	Johnson	N/A	N/A
2013/0045753	12/2012	Obermeyer et al.	N/A	N/A
2013/0050260	12/2012	Reitan	N/A	N/A
2013/0055083	12/2012	Fino	N/A	N/A
2013/0057587	12/2012	Leonard et al.	N/A	N/A
2013/0059607	12/2012	Herz et al.	N/A	N/A
2013/0060690	12/2012	Oskolkov et al.	N/A	N/A
2013/0063369	12/2012	Malhotra et al.	N/A	N/A
2013/0067027	12/2012	Song et al.	N/A	N/A
2013/0071093	12/2012	Hanks et al.	N/A	N/A
2013/0073970	12/2012	Piantino et al.	N/A	N/A
2013/0073984	12/2012	Lessin et al.	N/A	N/A
2013/0077887	12/2012	Elton	382/264	H04N 5/2628
2013/0080254	12/2012	Thramann	N/A	N/A
2013/0085790	12/2012	Palmer et al.	N/A	N/A
2013/0086072	12/2012	Peng et al.	N/A	N/A
2013/0090171	12/2012	Holton et al.	N/A	N/A
2013/0095857	12/2012	Garcia et al.	N/A	N/A
2013/0103760	12/2012	Golding et al.	N/A	N/A
2013/0103766	12/2012	Gupta	N/A	N/A
2013/0104053	12/2012	Thornton et al.	N/A	N/A
2013/0110631	12/2012	Mitchell et al.	N/A	N/A
2013/0110885	12/2012	Brundrett, III	N/A	N/A
2013/0111354	12/2012	Marra et al.	N/A	N/A
2013/0111514	12/2012	Slavin et al.	N/A	N/A

2013/0124091	12/2012	Matas et al.	N/A	N/A
2013/0128059	12/2012	Kristensson	N/A	N/A
2013/0129084	12/2012	Appleton	N/A	N/A
2013/0129252	12/2012	Lauper	N/A	N/A
2013/0132477	12/2012	Bosworth et al.	N/A	N/A
2013/0141463	12/2012	Barnett et al.	N/A	N/A
2013/0145286	12/2012	Feng et al.	N/A	N/A
2013/0151988	12/2012	Sorin et al.	N/A	N/A
2013/0152000	12/2012	Liu et al.	N/A	N/A
2013/0155169	12/2012	Hoover et al.	N/A	N/A
2013/0159110	12/2012	Rajaram et al.	N/A	N/A
2013/0159919	12/2012	Leydon	N/A	N/A
2013/0169822	12/2012	Zhu et al.	N/A	N/A
2013/0173729	12/2012	Starenky et al.	N/A	N/A
2013/0179520	12/2012	Lee et al.	N/A	N/A
2013/0182133	12/2012	Tanabe	N/A	N/A
2013/0185131	12/2012	Sinha et al.	N/A	N/A
2013/0191198	12/2012	Carlson et al.	N/A	N/A
2013/0194301	12/2012	Robbins et al.	N/A	N/A
2013/0198176	12/2012	Kim	N/A	N/A
2013/0201187	12/2012	Tong et al.	N/A	N/A
2013/0218965	12/2012	Abrol et al.	N/A	N/A
2013/0218968	12/2012	Mcevilly et al.	N/A	N/A
2013/0222323	12/2012	Mckenzie	N/A	N/A
2013/0227476	12/2012	Frey	N/A	N/A
2013/0232194	12/2012	Knapp et al.	N/A	N/A
2013/0249948	12/2012	Reitan	N/A	N/A
2013/0257877	12/2012	Davis	N/A	N/A
2013/0258040	12/2012	Kaytaz et al.	N/A	N/A
2013/0258117	12/2012	Penov	348/207.1	G06V 10/235
2013/0260800	12/2012	Asakawa et al.	N/A	N/A
2013/0263031	12/2012	Oshiro et al.	N/A	N/A
2013/0265450	12/2012	Barnes, Jr.	N/A	N/A
2013/0267253	12/2012	Case et al.	N/A	N/A
2013/0275505	12/2012	Gauglitz et al.	N/A	N/A
2013/0290443	12/2012	Collins et al.	N/A	N/A
2013/0304646	12/2012	De Geer	N/A	N/A
2013/0311255	12/2012	Cummins et al.	N/A	N/A
2013/0311452	12/2012	Jacoby	N/A	N/A
2013/0325964	12/2012	Berberat	N/A	N/A
2013/0332068	12/2012	Kesar et al.	N/A	N/A
2013/0339868	12/2012	Sharpe et al.	N/A	N/A
2013/0344896	12/2012	Kirmse et al.	N/A	N/A
2013/0346869	12/2012	Asver et al.	N/A	N/A
2013/0346877	12/2012	Borovoy et al.	N/A	N/A
2014/0006129	12/2013	Heath	N/A	N/A
2014/0011538	12/2013	Mulcahy et al.	N/A	N/A
2014/0011576	12/2013	Barbalet et al.	N/A	N/A
2014/0019264	12/2013	Wachman et al.	N/A	N/A

2014/0032682	12/2013	Prado et al.	N/A	N/A
2014/0035913	12/2013	Higgins	345/420	G06Q 30/0641
2014/0040066	12/2013	Fujioka	N/A	N/A
2014/0043204	12/2013	Basnayake et al.	N/A	N/A
2014/0043329	12/2013	Wang et al.	N/A	N/A
2014/0044368	12/2013	Tanner	382/236	G06T 7/207
2014/0045530	12/2013	Gordon et al.	N/A	N/A
2014/0047016	12/2013	Rao	N/A	N/A
2014/0047045	12/2013	Baldwin et al.	N/A	N/A
2014/0047335	12/2013	Lewis et al.	N/A	N/A
2014/0049652	12/2013	Moon et al.	N/A	N/A
2014/0052485	12/2013	Shidfar	N/A	N/A
2014/0052633	12/2013	Gandhi	N/A	N/A
2014/0055554	12/2013	Du et al.	N/A	N/A
2014/0057660	12/2013	Wager	N/A	N/A
2014/0082651	12/2013	Sharifi	N/A	N/A
2014/0085293	12/2013	Konoplev et al.	N/A	N/A
2014/0092130	12/2013	Anderson et al.	N/A	N/A
2014/0095293	12/2013	Abhyanker	N/A	N/A
2014/0096029	12/2013	Schultz	N/A	N/A
2014/0114565	12/2013	Aziz et al.	N/A	N/A
2014/0122658	12/2013	Haeger et al.	N/A	N/A
2014/0122787	12/2013	Shalvi et al.	N/A	N/A
2014/0125678	12/2013	Wang et al.	N/A	N/A
2014/0128166	12/2013	Tam et al.	N/A	N/A
2014/0129343	12/2013	Finster et al.	N/A	N/A
2014/0129953	12/2013	Spiegel	N/A	N/A
2014/0143143	12/2013	Fasoli et al.	N/A	N/A
2014/0149519	12/2013	Redfern et al.	N/A	N/A
2014/0155102	12/2013	Cooper et al.	N/A	N/A
2014/0157139	12/2013	Coroy et al.	N/A	N/A
2014/0160149	12/2013	Blackstock et al.	N/A	N/A
2014/0173424	12/2013	Hogeg et al.	N/A	N/A
2014/0173457	12/2013	Wang et al.	N/A	N/A
2014/0176662	12/2013	Goodman	N/A	N/A
2014/0189592	12/2013	Benchenaa et al.	N/A	N/A
2014/0199970	12/2013	Klotz	N/A	N/A
2014/0201527	12/2013	Krivorot	N/A	N/A
2014/0207679	12/2013	Cho	N/A	N/A
2014/0214471	12/2013	Schreiner, III	N/A	N/A
2014/0218394	12/2013	Hochmuth et al.	N/A	N/A
2014/0221089	12/2013	Fortkort	N/A	N/A
2014/0222564	12/2013	Kranendonk et al.	N/A	N/A
2014/0223372	12/2013	Dostie et al.	N/A	N/A
2014/0258405	12/2013	Perkin	N/A	N/A
2014/0265359	12/2013	Cheng et al.	N/A	N/A
2014/0266703	12/2013	Dalley, Jr. et al.	N/A	N/A
2014/0279061	12/2013	Elimeliah et al.	N/A	N/A
2014/0279436	12/2013	Dorsey et al.	N/A	N/A

2014/0279540	12/2013	Jackson	N/A	N/A
2014/0280058	12/2013	St. Clair	N/A	N/A
2014/0280537	12/2013	Pridmore et al.	N/A	N/A
2014/0282096	12/2013	Rubinstein et al.	N/A	N/A
2014/0287779	12/2013	O'keefe et al.	N/A	N/A
2014/0289216	12/2013	Voellmer et al.	N/A	N/A
2014/0289833	12/2013	Briceno	N/A	N/A
2014/0306884	12/2013	Sano et al.	N/A	N/A
2014/0306986	12/2013	Gottesman et al.	N/A	N/A
2014/0317302	12/2013	Naik	N/A	N/A
2014/0324627	12/2013	Haver et al.	N/A	N/A
2014/0324629	12/2013	Jacobs	N/A	N/A
2014/0325383	12/2013	Brown et al.	N/A	N/A
2014/0347368	12/2013	Kishore et al.	N/A	N/A
2014/0359024	12/2013	Spiegel	N/A	N/A
2014/0359032	12/2013	Spiegel et al.	N/A	N/A
2014/0362091	12/2013	Bouaziz et al.	N/A	N/A
2014/0372420	12/2013	Slep	N/A	N/A
2014/0380195	12/2013	Graham et al.	N/A	N/A
2015/0020086	12/2014	Chen et al.	N/A	N/A
2015/0046278	12/2014	Pei et al.	N/A	N/A
2015/0067880	12/2014	Ward et al.	N/A	N/A
2015/0071619	12/2014	Brough	N/A	N/A
2015/0084984	12/2014	Tomii et al.	N/A	N/A
2015/0086087	12/2014	Ricanek, Jr. et al.	N/A	N/A
2015/0087263	12/2014	Branscomb et al.	N/A	N/A
2015/0088464	12/2014	Yuen et al.	N/A	N/A
2015/0088622	12/2014	Ganschow et al.	N/A	N/A
2015/0095020	12/2014	Leydon	N/A	N/A
2015/0096042	12/2014	Mizrachi	N/A	N/A
2015/0116529	12/2014	Wu et al.	N/A	N/A
2015/0121251	12/2014	Kadirvel et al.	N/A	N/A
2015/0123967	12/2014	Quinn	345/419	G06T 7/60
2015/0160832	12/2014	Walkin et al.	N/A	N/A
2015/0169139	12/2014	Leva et al.	N/A	N/A
2015/0169142	12/2014	Longo et al.	N/A	N/A
2015/0169827	12/2014	Laborde	N/A	N/A
2015/0169938	12/2014	Yao et al.	N/A	N/A
2015/0172393	12/2014	Oplinger et al.	N/A	N/A
2015/0172534	12/2014	Miyakawa et al.	N/A	N/A
2015/0178260	12/2014	Brunson	N/A	N/A
2015/0181380	12/2014	Altman et al.	N/A	N/A
2015/0193522	12/2014	Choi et al.	N/A	N/A
2015/0193585	12/2014	Sunna	N/A	N/A
2015/0193819	12/2014	Chang	N/A	N/A
2015/0195235	12/2014	Trussel et al.	N/A	N/A
2015/0199082	12/2014	Scholler et al.	N/A	N/A
2015/0201030	12/2014	Longo et al.	N/A	N/A
2015/0206349	12/2014	Rosenthal et al.	N/A	N/A
2015/0213604	12/2014	Li et al.	N/A	N/A

2015/0220774	12/2014	Ebersman et al.	N/A	N/A
2015/0222814	12/2014	Li et al.	N/A	N/A
2015/0227602	12/2014	Ramu et al.	N/A	N/A
2015/0232065	12/2014	Ricci et al.	N/A	N/A
2015/0234942	12/2014	Harmon	N/A	N/A
2015/0245168	12/2014	Martin	N/A	N/A
2015/0261917	12/2014	Smith	N/A	N/A
2015/0264432	12/2014	Cheng	N/A	N/A
2015/0279098	12/2014	Kim et al.	N/A	N/A
2015/0295866	12/2014	Collet et al.	N/A	N/A
2015/0304806	12/2014	Vincent	N/A	N/A
2015/0312184	12/2014	Langholz et al.	N/A	N/A
2015/0334077	12/2014	Feldman	N/A	N/A
2015/0347519	12/2014	Hornkvist et al.	N/A	N/A
2015/0350136	12/2014	Flynn, III et al.	N/A	N/A
2015/0350262	12/2014	Rainisto et al.	N/A	N/A
2015/0365795	12/2014	Allen et al.	N/A	N/A
2015/0369623	12/2014	Blumenberg et al.	N/A	N/A
2015/0378502	12/2014	Hu et al.	N/A	N/A
2016/0006927	12/2015	Sehn	N/A	N/A
2016/0014063	12/2015	Hogeg et al.	N/A	N/A
2016/0021153	12/2015	Hull et al.	N/A	N/A
2016/0042548	12/2015	Du	345/473	G06T 13/40
2016/0045834	12/2015	Burns	N/A	N/A
2016/0050169	12/2015	Ben et al.	N/A	N/A
2016/0078095	12/2015	Man et al.	N/A	N/A
2016/0085773	12/2015	Chang et al.	N/A	N/A
2016/0085863	12/2015	Allen et al.	N/A	N/A
2016/0086500	12/2015	Kaleal, III	N/A	N/A
2016/0086670	12/2015	Gross et al.	N/A	N/A
2016/0093078	12/2015	Davis et al.	N/A	N/A
2016/0099901	12/2015	Allen et al.	N/A	N/A
2016/0110922	12/2015	Haring	N/A	N/A
2016/0134840	12/2015	Mcculloch	N/A	N/A
2016/0158600	12/2015	Rolley	N/A	N/A
2016/0163084	12/2015	Corazza et al.	N/A	N/A
2016/0164823	12/2015	Nordstrom et al.	N/A	N/A
2016/0179297	12/2015	Lundin et al.	N/A	N/A
2016/0180391	12/2015	Zabaneh	N/A	N/A
2016/0180447	12/2015	Kamalie et al.	N/A	N/A
2016/0180887	12/2015	Sehn	N/A	N/A
2016/0182422	12/2015	Sehn et al.	N/A	N/A
2016/0182875	12/2015	Sehn	N/A	N/A
2016/0188997	12/2015	Desnoyer et al.	N/A	N/A
2016/0189310	12/2015	O'kane	N/A	N/A
2016/0210500	12/2015	Feng et al.	N/A	N/A
2016/0217292	12/2015	Faaborg et al.	N/A	N/A
2016/0234149	12/2015	Tsuda et al.	N/A	N/A
2016/0239248	12/2015	Sehn	N/A	N/A
2016/0241504	12/2015	Raji et al.	N/A	N/A

2016/0253807	12/2015	Jones et al.	N/A	N/A
2016/0275721	12/2015	Park et al.	N/A	N/A
2016/0277419	12/2015	Allen et al.	N/A	N/A
2016/0286244	12/2015	Chang et al.	N/A	N/A
2016/0292273	12/2015	Murphy et al.	N/A	N/A
2016/0292905	12/2015	Nehmadi et al.	N/A	N/A
2016/0294891	12/2015	Miller	N/A	N/A
2016/0314759	12/2015	Shin et al.	N/A	N/A
2016/0321708	12/2015	Sehn	N/A	N/A
2016/0343160	12/2015	Blattner et al.	N/A	N/A
2016/0350297	12/2015	Riza	N/A	N/A
2016/0357578	12/2015	Kim et al.	N/A	N/A
2016/0359957	12/2015	Laliberte	N/A	N/A
2016/0359987	12/2015	Laliberte	N/A	N/A
2016/0359993	12/2015	Hendrickson et al.	N/A	N/A
2016/0378278	12/2015	Sirpal	N/A	N/A
2016/0379415	12/2015	Espeset et al.	N/A	N/A
2017/0006094	12/2016	Abou Mahmoud et al.	N/A	N/A
2017/0006322	12/2016	Dury et al.	N/A	N/A
2017/0027528	12/2016	Kaleal, III et al.	N/A	N/A
2017/0034173	12/2016	Miller et al.	N/A	N/A
2017/0039752	12/2016	Quinn et al.	N/A	N/A
2017/0061308	12/2016	Chen et al.	N/A	N/A
2017/0064240	12/2016	Mangat et al.	N/A	N/A
2017/0069124	12/2016	Tong et al.	N/A	N/A
2017/0076011	12/2016	Gannon	N/A	N/A
2017/0080346	12/2016	Abbas	N/A	N/A
2017/0087473	12/2016	Siegel et al.	N/A	N/A
2017/0113140	12/2016	Blackstock et al.	N/A	N/A
2017/0118145	12/2016	Aittoniemi et al.	N/A	N/A
2017/0124116	12/2016	League	N/A	N/A
2017/0126592	12/2016	El Ghouli	N/A	N/A
2017/0132649	12/2016	Oliva et al.	N/A	N/A
2017/0161382	12/2016	Ouimet et al.	N/A	N/A
2017/0199855	12/2016	Fishbeck	N/A	N/A
2017/0235848	12/2016	Van Dusen et al.	N/A	N/A
2017/0256039	12/2016	Hsu et al.	N/A	N/A
2017/0263029	12/2016	Yan et al.	N/A	N/A
2017/0270970	12/2016	Ho et al.	N/A	N/A
2017/0286752	12/2016	Gusarov et al.	N/A	N/A
2017/0287006	12/2016	Azmoodeh et al.	N/A	N/A
2017/0293673	12/2016	Purumala et al.	N/A	N/A
2017/0295250	12/2016	Samaranayake et al.	N/A	N/A
2017/0310934	12/2016	Du et al.	N/A	N/A
2017/0312634	12/2016	Ledoux et al.	N/A	N/A
2017/0324688	12/2016	Collet et al.	N/A	N/A
2017/0336960	12/2016	Chaudhri et al.	N/A	N/A
2017/0339006	12/2016	Austin et al.	N/A	N/A
2017/0353477	12/2016	Faigon et al.	N/A	N/A

2017/0358117	12/2016	Goossens	N/A	H04L 51/10
2017/0374003	12/2016	Allen et al.	N/A	N/A
2017/0374508	12/2016	Davis et al.	N/A	N/A
2018/0005420	12/2017	Bondich et al.	N/A	N/A
2018/0024726	12/2017	Hviding	N/A	N/A
2018/0025367	12/2017	Jain	N/A	N/A
2018/0032212	12/2017	Choi et al.	N/A	N/A
2018/0047200	12/2017	O'hara et al.	N/A	N/A
2018/0060363	12/2017	Ko et al.	N/A	N/A
2018/0068019	12/2017	Novikoff et al.	N/A	N/A
2018/0088777	12/2017	Daze et al.	N/A	N/A
2018/0091732	12/2017	Wilson et al.	N/A	N/A
2018/0097762	12/2017	Garcia et al.	N/A	N/A
2018/0113587	12/2017	Allen et al.	N/A	N/A
2018/0115503	12/2017	Baldwin et al.	N/A	N/A
2018/0315076	12/2017	Andreou	N/A	N/A
2018/0315133	12/2017	Brody et al.	N/A	N/A
2018/0315134	12/2017	Amitay et al.	N/A	N/A
2018/0374242	12/2017	Li et al.	N/A	N/A
2019/0001223	12/2018	Blackstock et al.	N/A	N/A
2019/0057616	12/2018	Cohen et al.	N/A	N/A
2019/0097958	12/2018	Collet et al.	N/A	N/A
2019/0188920	12/2018	Mcphee et al.	N/A	N/A
2019/0220932	12/2018	Amitay et al.	N/A	N/A
2019/0266390	12/2018	Gusarov et al.	N/A	N/A
2019/0287287	12/2018	Bondich et al.	N/A	N/A
2019/0386941	12/2018	Baldwin et al.	N/A	N/A
2020/0117339	12/2019	Amitay et al.	N/A	N/A
2020/0117340	12/2019	Amitay et al.	N/A	N/A
2020/0120097	12/2019	Amitay et al.	N/A	N/A
2020/0120170	12/2019	Amitay et al.	N/A	N/A
2020/0226848	12/2019	Van Os	N/A	G06T 11/001
2021/0266277	12/2020	Allen et al.	N/A	N/A
2023/0269208	12/2022	Allen et al.	N/A	N/A
2024/0195767	12/2023	Allen et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2009283063	12/2009	AU	N/A
2887596	12/2014	CA	N/A
101098241	12/2007	CN	N/A
102016887	12/2010	CN	N/A
102289796	12/2010	CN	N/A
102609247	12/2011	CN	N/A
103426194	12/2012	CN	N/A
103635892	12/2013	CN	N/A
105118023	12/2014	CN	N/A
105190700	12/2014	CN	N/A
105279354	12/2015	CN	N/A

105765620	12/2015	CN	N/A
108885795	12/2017	CN	N/A
109643370	12/2018	CN	N/A
109863532	12/2018	CN	N/A
110023985	12/2018	CN	N/A
110168478	12/2018	CN	N/A
110799937	12/2019	CN	N/A
110800018	12/2019	CN	N/A
110832538	12/2019	CN	N/A
110945555	12/2019	CN	N/A
111010882	12/2019	CN	N/A
111343075	12/2019	CN	N/A
111489264	12/2019	CN	N/A
109863532	12/2023	CN	N/A
117749754	12/2023	CN	N/A
2051480	12/2008	EP	N/A
2151797	12/2009	EP	N/A
2184092	12/2009	EP	N/A
2399928	12/2003	GB	N/A
2001230801	12/2000	JP	N/A
2014006881	12/2013	JP	N/A
549793182	12/2013	JP	N/A
19990073076	12/1998	KR	N/A
20000063919	12/1999	KR	N/A
20010078417	12/2000	KR	N/A
20040063436	12/2003	KR	N/A
1020050036963	12/2004	KR	N/A
20070008417	12/2006	KR	N/A
1020120070898	12/2011	KR	N/A
20130069730	12/2012	KR	N/A
20140033088	12/2013	KR	N/A
101445263	12/2013	KR	N/A
20140107253	12/2013	KR	N/A
20160016532	12/2015	KR	N/A
20160025365	12/2015	KR	N/A
20160028028	12/2015	KR	N/A
20160051536	12/2015	KR	N/A
20170091803	12/2016	KR	N/A
102253891	12/2020	KR	N/A
102372756	12/2021	KR	N/A
102530504	12/2022	KR	N/A
102649278	12/2023	KR	N/A
102720966	12/2023	KR	N/A
WO-1996024213	12/1995	WO	N/A
WO-1999063453	12/1998	WO	N/A
WO-2000058882	12/1999	WO	N/A
WO-2001029642	12/2000	WO	N/A
WO-2001050703	12/2000	WO	N/A
WO-2003094072	12/2002	WO	N/A
WO-2004095308	12/2003	WO	N/A

WO-2006107182	12/2005	WO	N/A
WO-2006118755	12/2005	WO	N/A
WO-2007092668	12/2006	WO	N/A
WO-2007134402	12/2006	WO	N/A
WO-2009043020	12/2008	WO	N/A
WO-2011040821	12/2010	WO	N/A
WO-2011119407	12/2010	WO	N/A
WO-2012000107	12/2011	WO	N/A
WO-2012139276	12/2011	WO	N/A
WO-2013008238	12/2012	WO	N/A
WO-2013008251	12/2012	WO	N/A
WO-2013027893	12/2012	WO	N/A
WO-2013045753	12/2012	WO	N/A
WO-2013152454	12/2012	WO	N/A
WO-2013166588	12/2012	WO	N/A
WO-2014006129	12/2013	WO	N/A
WO-2014031899	12/2013	WO	N/A
WO-2014068573	12/2013	WO	N/A
WO-2014115136	12/2013	WO	N/A
WO-2014194262	12/2013	WO	N/A
WO-2014194439	12/2013	WO	N/A
WO-2015192026	12/2014	WO	N/A
WO-2016044424	12/2015	WO	N/A
WO-2016054562	12/2015	WO	N/A
WO-2016065131	12/2015	WO	N/A
WO-2016/100342	12/2015	WO	N/A
WO-2016090605	12/2015	WO	N/A
WO-2016100318	12/2015	WO	N/A
WO-2016100318	12/2015	WO	N/A
WO-2016/112299	12/2015	WO	N/A
WO-2016149594	12/2015	WO	N/A
WO-2016179166	12/2015	WO	N/A
WO-2016179235	12/2015	WO	N/A
WO-2017173319	12/2016	WO	N/A
WO-2017176739	12/2016	WO	N/A
WO-2017176992	12/2016	WO	N/A
WO-2018005644	12/2017	WO	N/A
WO-2018006053	12/2017	WO	N/A
WO-2018081013	12/2017	WO	N/A
WO-2018102562	12/2017	WO	N/A
WO-2018129531	12/2017	WO	N/A
WO-2018200042	12/2017	WO	N/A
WO-2018200043	12/2017	WO	N/A
WO-2018201102	12/2017	WO	N/A
WO-2018201104	12/2017	WO	N/A
WO-2018201106	12/2017	WO	N/A
WO-2018201107	12/2017	WO	N/A
WO-2018201108	12/2017	WO	N/A
WO-2018201109	12/2017	WO	N/A
WO-2019089613	12/2018	WO	N/A

OTHER PUBLICATIONS

Xu, B., Chang, P., Welker, C. L., Bazarova, N. N., & Cosley, D. (Feb. 2016). Automatic archiving versus default deletion: what Snapchat tells us about ephemerality in design. In Proceedings of the 19th ACM conference on computer-supported cooperative work & social computing (pp. 1662-1675). cited by examiner

Frisch, M., Kleinau, S., Langner, R., & Dachsel, R. (May 2011). Grids & guides: multi-touch layout and alignment tools. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (pp. 1615-1618). cited by examiner

"A Guide to Advertising on Campus With Snapchat Geofilters", College Marketing Group, [Online] Retrieved from the Internet: <URL: <https://collegemarketinggroup.com/blog/a-guide-toadvertising-on-campus-with-snapchat-geofilters/>>, (Jul. 25, 2016), 5 pgs. cited by applicant

"A Whole New Story", Snap, Inc., [Online] Retrieved from the Internet: <URL: <https://www.snap.com/en-us/news/>>, (2017), 13 pgs. cited by applicant

"Adding photos to your listing", eBay, [Online] Retrieved from the Internet: <URL: <http://pages.ebay.com/help/sell/pictures.html>>, (accessed May 24, 2017), 4 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Corrected Notice of Allowability mailed Jan. 28, 2021", 3 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Advisory Action mailed Mar. 11, 2020", 2 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Final Office Action mailed Feb. 12, 2021", 10 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Final Office Action mailed Nov. 21, 2019", 9 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Non Final Office Action mailed Mar. 30, 2020", 9 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Non Final Office Action mailed Aug. 6, 2019", 9 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Non Final Office Action mailed Oct. 27, 2020", 10 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Response filed Jan. 27, 2021 to Non Final Office Action mailed Oct. 27, 2020", 9 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Response filed Feb. 21, 2020 to Final Office Action mailed Nov. 21, 2019", 9 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Response filed Jul. 30, 2020 to Non Final Office Action mailed Mar. 30, 2020", 10 pgs. cited by applicant

"U.S. Appl. No. 15/401,926, Response filed Nov. 6, 2019 to Non Final Office Action mailed Aug. 6, 2019", 10 pgs. cited by applicant

"U.S. Appl. No. 15/583,142, Notice of Allowance mailed Jun. 6, 2019", 8 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Final Office Action mailed Apr. 13, 2020", 45 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Final Office Action mailed Jun. 10, 2019", 44 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Non Final Office Action mailed Oct. 30, 2019", 45 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Notice of Allowance mailed Sep. 29, 2020", 13 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Response filed Jan. 30, 2020 to Non Final Office Action mailed Oct. 30, 2019", 17 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Response filed Jul. 13, 2020 to Final Office Action mailed Apr. 13, 2020", 20 pgs. cited by applicant

"U.S. Appl. No. 15/628,408, Response filed Aug. 12, 2019 to Final Office Action mailed Jun. 10, 2019", 12 pgs. cited by applicant

"U.S. Appl. No. 15/859,101, Examiner Interview Summary mailed Sep. 18, 2018", 3 pgs. cited by applicant

"U.S. Appl. No. 15/859,101, Non Final Office Action mailed Jun. 15, 2018", 10 pgs. cited by applicant

"U.S. Appl. No. 15/859,101, Notice of Allowance mailed Oct. 4, 2018", 9 pgs. cited by applicant

"U.S. Appl. No. 15/859,101, Response filed Sep. 17, 2018 to Non Final Office Action mailed Jun. 15, 2018", 17 pgs. cited by applicant

"U.S. Appl. No. 15/901,387, Non Final Office Action mailed Oct. 30, 2019", 40 pgs. cited by applicant

“U.S. Appl. No. 15/965,361, Non Final Office Action mailed Jun. 22, 2020”, 35 pgs. cited by applicant

“U.S. Appl. No. 15/965,744, Examiner Interview Summary mailed Feb. 21, 2020”, 3 pgs. cited by applicant

“U.S. Appl. No. 15/965,744, Final Office Action mailed Feb. 6, 2020”, 19 pgs. cited by applicant

“U.S. Appl. No. 15/965,744, Non Final Office Action mailed Feb. 1, 2021”, 29 pgs. cited by applicant

“U.S. Appl. No. 15/965,744, Non Final Office Action mailed Jun. 12, 2019”, 18 pgs. cited by applicant

“U.S. Appl. No. 15/965,744, Response filed Jun. 8, 2020 to Final Office Action mailed Feb. 6, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 15/965,744, Response filed Nov. 12, 2019 to Non Final Office Action mailed Jun. 12, 2019”, 10 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Examiner Interview Summary mailed Jul. 29, 2020”, 3 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Final Office Action mailed Jun. 11, 2020”, 12 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Non Final Office Action mailed Jan. 27, 2020”, 9 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Non Final Office Action mailed Jul. 10, 2019”, 8 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Non Final Office Action mailed Nov. 30, 2020”, 13 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Response filed Feb. 28, 2020 to Non Final Office Action mailed Jan. 27, 2020”, 12 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Response filed Oct. 10, 2019 to Non-Final Office Action mailed Jul. 10, 2019”, 11 pgs. cited by applicant

“U.S. Appl. No. 15/965,749, Response filed Oct. 12, 2020 to Final Office Action mailed Jun. 11, 2020”, 14 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Corrected Notice of Allowability mailed Jan. 6, 2021”, 2 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Corrected Notice of Allowability mailed Mar. 1, 2021”, 2 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Final Office Action mailed Jul. 17, 2020”, 14 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Non Final Office Action mailed Mar. 30, 2020”, 13 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Notice of Allowance mailed Nov. 16, 2020”, 7 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Response filed Jun. 30, 2020 to Non Final Office Action mailed Mar. 30, 2020”, 12 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Response filed Oct. 19, 2020 to Final Office Action mailed Jul. 17, 2020”, 14 pgs. cited by applicant

“U.S. Appl. No. 15/965,754, Supplemental Notice of Allowability mailed Dec. 16, 2020”, 2 pgs. cited by applicant

“U.S. Appl. No. 15/965,756, Non Final Office Action mailed Jan. 13, 2021”, 16 pgs. cited by applicant

“U.S. Appl. No. 15/965,756, Non Final Office Action mailed Jun. 24, 2020”, 16 pgs. cited by applicant

“U.S. Appl. No. 15/965,756, Response filed Sep. 24, 2020 to Non Final Office Action mailed Jun. 24, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 15/965,764, Examiner Interview Summary mailed Aug. 6, 2020”, 3 pgs. cited by applicant

“U.S. Appl. No. 15/965,764, Non Final Office Action mailed Jan. 2, 2020”, 18 pgs. cited by applicant

“U.S. Appl. No. 15/965,764, Final Office Action mailed May 14, 2020”, 18 pgs. cited by applicant

“U.S. Appl. No. 15/965,764, Non Final Office Action mailed Feb. 22, 2021”, 18 pgs. cited by applicant

“U.S. Appl. No. 15/965,764, Response filed Apr. 2, 2020 to Non Final Office Action mailed Jan. 2, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 15/965,764, Response filed Oct. 14, 2020 to Final Office Action mailed May 14, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Final Office Action mailed Jan. 30, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Non Final Office Action mailed Jun. 19, 2020”, 12 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Non Final Office Action mailed Jul. 29, 2019”, 8 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Non Final Office Action mailed Oct. 16, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Response filed Mar. 16, 2021 to Non Final Office Action mailed Oct. 16, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Response filed Jun. 1, 2020 to Final Office Action mailed Jan. 30, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Response filed Jul. 7, 2020 to Non Final Office Action mailed Jun. 19, 2020”, 9 pgs. cited by applicant

“U.S. Appl. No. 15/965,775, Response filed Oct. 29, 2019 to Non Final Office Action mailed Jul. 29, 2019”, 10 pgs. cited by applicant

“U.S. Appl. No. 15/965,811, Final Office Action mailed Feb. 12, 2020”, 16 pgs. cited by applicant

“U.S. Appl. No. 15/965,811, Non Final Office Action mailed Jun. 26, 2020”, 20 pgs. cited by applicant

“U.S. Appl. No. 15/965,811, Non Final Office Action mailed Aug. 8, 2019”, 15 pgs. cited by applicant

“U.S. Appl. No. 15/965,811, Response filed Jun. 12, 2020 to Final Office Action mailed Feb. 12, 2020”, 13 pgs. cited by applicant

“U.S. Appl. No. 15/965,811, Response filed Nov. 8, 2019 to Non Final Office Action mailed Aug. 8, 2019”, 14 pgs. cited by applicant

“U.S. Appl. No. 16/115,259, Final Office Action mailed Jul. 13, 2021”, 18 pgs. cited by applicant

“U.S. Appl. No. 16/115,259, Non Final Office Action mailed Jan. 11, 2021”, 17 pgs. cited by applicant

“U.S. Appl. No. 16/115,259, Response filed May 11, 2021 to Non Final Office Action mailed Jan. 11, 2021”, 14 pgs. cited by applicant

“U.S. Appl. No. 16/115,259, Response filed Oct. 13, 2021 to Final Office Action mailed Jul. 13, 2021”, 10 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Final Office Action mailed Feb. 8, 2021”, 8 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Final Office Action mailed Jul. 7, 2020”, 8 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Non Final Office Action mailed Feb. 5, 2020”, 7 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Non Final Office Action mailed Oct. 30, 2020”, 9 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Response filed Feb. 1, 2021 to Non Final Office Action mailed Oct. 30, 2020”, 9 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Response filed May 5, 2020 to Non Final Office Action mailed Feb. 5, 2020”, 8 pgs. cited by applicant

“U.S. Appl. No. 16/126,869, Response filed Oct. 7, 2020 to Final Office Action mailed Jul. 7, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 16/193,938, Final Office Action mailed Aug. 28, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 16/193,938, Non Final Office Action mailed Jan. 16, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 16/193,938, Non Final Office Action mailed Feb. 24, 2021”, 10 pgs. cited by applicant

“U.S. Appl. No. 16/193,938, Response filed Mar. 24, 2020 to Non Final Office Action mailed Jan. 16, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 16/193,938, Response filed Nov. 30, 2020 to Final Office Action mailed Aug. 28, 2020”, 9 pgs. cited by applicant

“U.S. Appl. No. 16/232,824, Examiner Interview Summary mailed Jul. 24, 2020”, 3 pgs. cited by applicant

“U.S. Appl. No. 16/232,824, Final Office Action mailed Apr. 30, 2020”, 19 pgs. cited by applicant

“U.S. Appl. No. 16/232,824, Non Final Office Action mailed Feb. 19, 2021”, 28 pgs. cited by applicant

“U.S. Appl. No. 16/232,824, Non Final Office Action mailed Oct. 21, 2019”, 18 pgs. cited by applicant

“U.S. Appl. No. 16/232,824, Response filed Feb. 21, 2020 to Non Final Office Action mailed Oct. 21, 2019”, 9 pgs. cited by applicant

“U.S. Appl. No. 16/232,824, Response filed Jul. 15, 2020 to Final Office Action mailed Apr. 30, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 16/245,660, Final Office Action mailed Feb. 6, 2020”, 12 pgs. cited by applicant

“U.S. Appl. No. 16/245,660, Non Final Office Action mailed Jun. 27, 2019”, 11 pgs. cited by applicant
“U.S. Appl. No. 16/245,660, Notice of Allowability mailed Nov. 18, 2020”, 2 pgs. cited by applicant
“U.S. Appl. No. 16/245,660, Notice of Allowance mailed Jul. 8, 2020”, 8 pgs. cited by applicant
“U.S. Appl. No. 16/245,660, Notice of Allowance mailed Nov. 3, 2020”, 8 pgs. cited by applicant
“U.S. Appl. No. 16/245,660, Response filed Jun. 8, 2020 to Final Office Action mailed Feb. 6, 2020”, 16 pgs. cited by applicant
“U.S. Appl. No. 16/245,660, Response filed Nov. 6, 2019 to Non Final Office Action mailed Jun. 27, 2019”, 11 pgs. cited by applicant
“U.S. Appl. No. 16/365,300, Final Office Action mailed May 13, 2020”, 44 pgs. cited by applicant
“U.S. Appl. No. 16/365,300, Non Final Office Action mailed Sep. 28, 2020”, 40 pgs. cited by applicant
“U.S. Appl. No. 16/365,300, Non Final Office Action mailed Oct. 30, 2019”, 40 pgs. cited by applicant
“U.S. Appl. No. 16/365,300, Response filed Jan. 28, 2021 to Non Final Office Action mailed Sep. 28, 2020”, 17 pgs. cited by applicant
“U.S. Appl. No. 16/365,300, Response filed Jan. 30, 2020 to Non Final Office Action mailed Oct. 30, 2019”, 16 pgs. cited by applicant
“U.S. Appl. No. 16/365,300, Response filed Aug. 13, 2020 to Final Office Action mailed May 13, 2020”, 16 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Final Office Action mailed Jun. 15, 2020”, 12 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Final Office Action mailed Dec. 23, 2020”, 15 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Non Final Office Action mailed Jan. 8, 2020”, 14 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Non Final Office Action mailed Sep. 11, 2020”, 15 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Notice of Allowance mailed Feb. 22, 2021”, 7 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Response filed Feb. 8, 2021 to Final Office Action mailed Dec. 23, 2020”, 11 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Response filed Apr. 2, 2020 to Non Final Office Action mailed Jan. 8, 2020”, 10 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Response filed Aug. 5, 2020 to Final Office Action mailed Jun. 15, 2020”, 11 pgs. cited by applicant
“U.S. Appl. No. 16/409,390, Response filed Dec. 8, 2020 to Non Final Office Action mailed Sep. 11, 2020”, 12 pgs. cited by applicant
“U.S. Appl. No. 16/552,003, Notice of Allowance mailed Aug. 27, 2020”, 15 pgs. cited by applicant
“U.S. Appl. No. 16/563,445, Final Office Action mailed Mar. 8, 2021”, 11 pgs. cited by applicant
“U.S. Appl. No. 16/563,445, Non Final Office Action mailed Sep. 29, 2020”, 11 pgs. cited by applicant
“U.S. Appl. No. 16/563,445, Response filed Jan. 29, 2021 to Non Final Office Action mailed Sep. 29, 2020”, 9 pgs. cited by applicant
“U.S. Appl. No. 17/247,169, Preliminary Amendment filed Feb. 2, 2021”, 7 pgs. cited by applicant
“BlogStomp”, StompSoftware, [Online] Retrieved from the Internet: <URL: <http://stompsoftware.com/blogstomp>>, (accessed May 24, 2017), 12 pgs. cited by applicant
“Cup Magic Starbucks Holiday Red Cups come to life with AR app”, Blast Radius, [Online] Retrieved from the Internet: <URL: <https://web.archive.org/web/20160711202454/http://www.blastradius.com/work/cup-magic>>, (2016), 7 pgs. cited by applicant
“Daily App: InstaPlace (iOS/Android): Give Pictures a Sense of Place”, TechPP, [Online] Retrieved from the Internet: <URL: <http://techpp.com/2013/02/15/instaplace-app-review>>, (2013), 13 pgs. cited by applicant
“European Application Serial No. 19206595.1, Extended European Search Report mailed Mar. 31, 2020”, 6 pgs. cited by applicant
“European Application Serial No. 17751497.3, Communication Pursuant to Article 94(3) EPC mailed Jun. 2, 2021”, 4 pgs. cited by applicant
“European Application Serial No. 17776809.0, Communication Pursuant to Article 94(3) EPC mailed

Dec. 9, 2019", 4 pgs. cited by applicant

"European Application Serial No. 17776809.0, Response filed Mar. 19, 2020 to Communication Pursuant to Article 94(3) EPC mailed Dec. 9, 2019", 25 pgs. cited by applicant

"European Application Serial No. 17876226.6, Communication Pursuant to Article 94(3) EPC mailed May 29, 2020", 5 pgs. cited by applicant

"European Application Serial No. 17876226.6, Extended European Search Report mailed Sep. 5, 2019", 10 pgs. cited by applicant

"European Application Serial No. 17876226.6, Response filed Mar. 30, 2020 to Extended European Search Report mailed Sep. 5, 2019", 22 pgs. cited by applicant

"European Application Serial No. 17876226.6, Response filed Oct. 2, 2020 to Communication Pursuant to Article 94(3) EPC mailed May 29, 2020", 22 pgs. cited by applicant

"European Application Serial No. 18789872.1, Communication Pursuant to Article 94(3) EPC mailed Aug. 11, 2020", 6 pgs. cited by applicant

"European Application Serial No. 18789872.1, Extended European Search Report mailed Jan. 2, 2020", 8 pgs. cited by applicant

"European Application Serial No. 18789872.1, Response filed Feb. 18, 2021 to Communication Pursuant to Article 94(3) EPC mailed Aug. 11, 2020", 15 pgs. cited by applicant

"European Application Serial No. 18790189.7, Communication Pursuant to Article 94(3) EPC mailed Jul. 30, 2020", 9 pgs. cited by applicant

"European Application Serial No. 18790189.7, Extended European Search Report mailed Jan. 2, 2020", 7 pgs. cited by applicant

"European Application Serial No. 18790189.7, Response filed Feb. 9, 2021 to Communication Pursuant to Article 94(3) EPC mailed Jul. 30, 2020", 11 pgs. cited by applicant

"European Application Serial No. 18790189.7, Response Filed Jul. 14, 2020 to Extended European Search Report mailed Jan. 2, 2020", 21 pgs. cited by applicant

"European Application Serial No. 18790319.0, Extended European Search Report mailed Feb. 12, 2020", 6 pgs. cited by applicant

"European Application Serial No. 18790319.0, Response filed Aug. 27, 2020 to Extended European Search Report mailed Feb. 12, 2020", 19 pgs. cited by applicant

"European Application Serial No. 18791363.7, Communication Pursuant to Article 94(3) EPC mailed Aug. 11, 2020", 9 pgs. cited by applicant

"European Application Serial No. 18791363.7, Extended European Search Report mailed Jan. 2, 2020", 8 pgs. cited by applicant

"European Application Serial No. 18791363.7, Response filed Jul. 14, 2020 to Extended European Search Report mailed Jan. 2, 2020", 31 pgs. cited by applicant

"European Application Serial No. 18791925.3, Extended European Search Report mailed Jan. 2, 2020", 6 pgs. cited by applicant

"European Application Serial No. 18791925.3, Response Filed Jul. 27, 2020 to Extended European Search Report mailed Jan. 2, 2020", 19 pgs. cited by applicant

"European Application Serial No. 19206595.1, Response filed Dec. 16, 2020 to Extended European Search Report mailed Mar. 31, 2020", 43 pgs. cited by applicant

"European Application Serial No. 19206610.8, Extended European Search Report mailed Feb. 12, 2020", 6 pgs. cited by applicant

"European Application Serial No. 19206610.8, Response filed Sep. 23, 2020 to Extended European Search Report mailed Feb. 12, 2020", 109 pgs. cited by applicant

"InstaPlace Photo App Tell The Whole Story", [Online] Retrieved from the Internet: <URL: youtu.be/uF_gFkg1hBM>, (Nov. 8, 2013), 113 pgs., 1:02 min. cited by applicant

"International Application Serial No. PCT/US2015/037251, International Search Report mailed Sep. 29, 2015", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2017/063981, International Preliminary Report on

Patentability mailed Jun. 13, 2019”, 10 pgs. cited by applicant

“International Application Serial No. PCT/US2018/000112, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 6 pgs. cited by applicant

“International Application Serial No. PCT/US2018/000113, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 6 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030039, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 6 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030041, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 5 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030041, International Search Report mailed Jul. 11, 2018”, 2 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030041, Written Opinion mailed Jul. 11, 2018”, 3 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030043, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 7 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030044, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 8 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030045, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 8 pgs. cited by applicant

“International Application Serial No. PCT/US2018/030046, International Preliminary Report on Patentability mailed Nov. 7, 2019”, 8 pgs. cited by applicant

“Introducing Google Latitude”, Google UK, [Online] Retrieved from the Internet: <URL: <https://www.youtube.com/watch?v=XecGMKqiA5A>>, [Retrieved on: Oct. 23, 2019], (Feb. 3, 2009), 1 pg. cited by applicant

“Introducing Snapchat Stories”, [Online] Retrieved from the Internet: <URL: <https://web.archive.org/web/20131026084921/https://www.youtube.com/watch?v=88Cu3yN-LIM>>, (Oct. 3, 2013), 92 pgs.; 00:47 min. cited by applicant

“Korean Application Serial No. 10-2018-7031055, Notice of Preliminary Rejection mailed Aug. 6, 2019”, w/ English Translation, 13 pgs. cited by applicant

“Korean Application Serial No. 10-2018-7031055, Office Action mailed Feb. 25, 2020”, w/ English Translation, 7 pgs. cited by applicant

“Korean Application Serial No. 10-2018-7031055, Response filed Mar. 27, 2020 to Office Action mailed Feb. 25, 2020”, w/ English claims, 24 pgs. cited by applicant

“Korean Application Serial No. 10-2018-7031055, Response filed Oct. 7, 2019 to Notice of Preliminary Rejection mailed Aug. 6, 2019”, w/ English Claims, 30 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7002736, Response filed Dec. 28, 2020 to Final Office Action mailed Nov. 26, 2020”, w/ English Claims, 16 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7014555, Notice of Preliminary Rejection mailed Jul. 20, 2020”, w/ English Translation, 12 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7014555, Response filed Oct. 6, 2020 to Notice of Preliminary Rejection mailed Jul. 20, 2020”, w/ English Claims, 27 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7018501, Final Office Action mailed Sep. 8, 2020”, w/ English translation, 9 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7018501, Notice of Preliminary Rejection mailed Apr. 16, 2020”, w/ English Translation, 20 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7018501, Response filed Jun. 16, 2020 to Notice of Preliminary Rejection mailed Apr. 16, 2020”, w/ English Claims, 17 pgs. cited by applicant

“Korean Application Serial No. 10-2019-7018501, Response filed Dec. 7, 2020 to Final Office Action mailed Sep. 8, 2020”, w/ English Claims, 27 pgs. cited by applicant

“Korean Application Serial No. 10-2020-7022773, Notice of Preliminary Rejection mailed Feb. 26,

2021”, w/ English Translation, 11 pgs. cited by applicant

“Korean Application Serial No. 10-2020-7022773, Notice of Preliminary Rejection mailed Aug. 23, 2020”, w/ English translation, 11 pgs. cited by applicant

“Korean Application Serial No. 10-2020-7022773, Response filed Oct. 19, 2020 to Notice of Preliminary Rejection mailed Aug. 23, 2020”, w/ English Claims, 26 pgs. cited by applicant

“Korean Application Serial No. 10-2020-7035136, Notice of Preliminary Rejection mailed Feb. 25, 2021”, w/ English Translation, 5 pgs. cited by applicant

“Korean Application Serial No. 10-2021-7010821, Notice of Preliminary Rejection mailed May 28, 2021”, w/ English Translation, 18 pgs. cited by applicant

“Korean Application Serial No. 10-2021-7010821, Response filed Jul. 28, 2021 to Notice of Preliminary Rejection mailed May 28, 2021”, w/ English Claims, 31 pgs. cited by applicant

“Macy's Believe-o-Magic”, [Online] Retrieved from the Internet: <URL:

<https://web.archive.org/web/20190422101854/https://www.youtube.com/watch?v=xvzRXy3J0Z0&feature=youtu.be>>, (Nov. 7, 2011), 102 pgs.; 00:51 min. cited by applicant

“Macy's Introduces Augmented Reality Experience in Stores across Country as Part of Its 2011 Believe Campaign”, Business Wire, [Online] Retrieved from the Internet: <URL:

<https://www.businesswire.com/news/home/20111102006759/en/Macys-Introduces-Augmented-Reality-Experience-Stores-Country>>, (Nov. 2, 2011), 6 pgs. cited by applicant

“Starbucks Cup Magic”, [Online] Retrieved from the Internet: <URL: <https://www.youtube.com/watch?v=RWwQXi9RGOw>>, (Nov. 8, 2011), 87 pgs.; 00:47 min. cited by applicant

“Starbucks Cup Magic for Valentine's Day”, [Online] Retrieved from the Internet: <URL:

<https://www.youtube.com/watch?v=8nvqOzjq10w>>, (Feb. 6, 2012), 88 pgs.; 00:45 min. cited by applicant

“Starbucks Holiday Red Cups Come to Life, Signaling the Return of the Merriest Season”, Business Wire, [Online] Retrieved from the Internet: <URL:

<http://www.businesswire.com/news/home/20111115005744/en/2479513/Starbucks-Holiday-Red-Cups-Life-Signaling-Return>>, (Nov. 15, 2011), 5 pgs. cited by applicant

“The One Million Tweet Map: Using Maptimize to Visualize Tweets in a World Map | PowerPoint Presentation”, fppt.com, [Online] Retrieved from the Internet: <URL:

<https://web.archive.org/web/20121103231906/http://www.freepower-point-templates.com/articles/the-one-million-tweet-mapusing-maptimize-to-visualize-tweets-in-a-world-map/>>, (Nov. 3, 2012), 6 pgs. cited by applicant

Alex, Heath, “What do Snapchat's emojis mean?—Understanding these emojis will turn you into a Snapchat pro”, Business Insider, [Online] Retrieved from the Internet : <URL:

<https://www.businessinsider.com/what-do-snapchats-emojismean-2016-5?international=true&r=US&IR=T>>, (May 28, 2016), 1 pg. cited by applicant

Carthy, Roi, “Dear All Photo Apps: Mobli Just Won Filters”, TechCrunch, [Online] Retrieved from the Internet: <URL: <https://techcrunch.com/2011/09/08/mobli-filters>>, (Sep. 8, 2011), 10 pgs. cited by applicant

Castelluccia, Claude, et al., “EphPub: Toward robust Ephemeral Publishing”, 19th IEEE International Conference on Network Protocols (ICNP), (Oct. 17, 2011), 18 pgs. cited by applicant

Fajman, “An Extensible Message Format for Message Disposition Notifications”, Request for Comments: 2298, National Institutes of Health, (Mar. 1998), 28 pgs. cited by applicant

Finn, Greg, “Miss Google Latitude? Google Plus With Location Sharing is Now a Suitable Alternative”, Cypress North, [Online] Retrieved from the Internet: <URL:

<https://cypressnorth.com/social-media/miss-google-latitude-google-location-sharing-now-suitable-alternative/>>, [Retrieved on: Oct. 24, 2019], (Nov. 27, 2013), 10 pgs. cited by applicant

Gundersen, Eric, “Foursquare Switches to MapBox Streets, Joins the OpenStreetMap Movement”, [Online] Retrieved from the Internet: <URL: <https://blog.mapbox.com/foursquare-switches-to-mapbox-streets-joins-the-openstreetmap-movement-29e6a17f4464>>, (Mar. 6, 2012), 4 pgs. cited by

applicant

Janthong, Isaranu, "Instaplace ready on Android Google Play store", Android App Review Thailand, [Online] Retrieved from the Internet: <URL: <http://www.android-free-app-review.com/2013/01/instaplace-android-google-play-store.html>>, (Jan. 23, 2013), 9 pgs. cited by

applicant

Karen, Tumbokon, "Snapchat Update: How to Add Bitmoji to Customizable Geofilters", International Business Times, [Online] Retrieved from the Internet: <URL: <https://www.ibtimes.com/snapchat-update-how-add-bitmoji-customizable-geofilters-2448152>>, (Nov. 18, 2016), 6 pgs. cited by applicant

Lapenna, Joe, "The Official Google Blog. Check in with Google Latitude", Google Blog, [Online] Retrieved from the Internet:

<<https://web.archive.org/web/20110201201006/https://googleblog.blogspot.com/2011/02/check-in-with-google-latitude.html>>, [Retrieved on: Oct. 23, 2019], (Feb. 1, 2011), 6 pgs. cited by applicant

Macleod, Duncan, "Macys Believe-o-Magic App", [Online] Retrieved from the Internet: <URL: <http://theinspirationroom.com/daily/2011/macys-believe-o-magic-app>>, (Nov. 14, 2011), 10 pgs. cited by applicant

Macleod, Duncan, "Starbucks Cup Magic Lets Merry", [Online] Retrieved from the Internet: <URL: <http://theinspirationroom.com/daily/2011/starbucks-cup-magic>>, (Nov. 12, 2011), 8 pgs. cited by applicant

Melanson, Mike, "This text message will self destruct in 60 seconds", [Online] Retrieved from the Internet: <URL:

http://readwrite.com/2011/02/11/this_text_message_will_self_destruct_in_60_seconds>, (Feb. 18, 2015), 4 pgs. cited by applicant

Neis, Pascal, "The OpenStreetMap Contributors Map aka Who's around me?", [Online] Retrieved from the Internet by the examiner on Jun. 5, 2019: <URL: <https://neis-one.org/2013/01/oooc/>>, (Jan. 6, 2013), 7 pgs. cited by applicant

Notopoulos, Katie, "A Guide to the New Snapchat Filters and Big Fonts", [Online] Retrieved from the Internet: <URL: https://www.buzzfeed.com/katienotopoulos/a-guide-to-the-new-snapchat-filters-and-big-fonts?utm_term=.bkQ9qVZWe#.nv58YXpkV>, (Dec. 22, 2013), 13 pgs. cited by applicant

Panzarino, Matthew, "Snapchat Adds Filters, A Replay Function and For Whatever Reason, Time, Temperature And Speed Overlays", TechCrunch, [Online] Retrieved from the Internet: <URL: <https://techcrunch.com/2013/12/20/snapchat-adds-filters-new-font-and-for-some-reason-time-temperature-and-speed-overlays/>>, (Dec. 20, 2013), 12 pgs. cited by applicant

Perez, Sarah, "Life 360, The Family Locator With More Users Than Foursquare, Raises A \$10 Million Series B", [Online] Retrieved from the Internet: <URL: <https://techcrunch.com/2013/07/10/life360-the-family-locator-with-more-users-than-foursquare-raises-10-million-series-b/>>, (Jul. 10, 2013), 2 pgs. cited by applicant

Sawers, Paul, "Snapchat for iOS Lets You Send Photos to Friends and Set How long They're Visible For", [Online] Retrieved from the Internet: <URL: <https://thenextweb.com/apps/2012/05/07/snapchat-for-ios-lets-you-send-photos-to-friends-and-set-how-long-theyre-visible-for/>>, (May 7, 2012), 5 pgs. cited by applicant

Shein, Esther, "Ephemeral Data", Communications of the ACM, vol. 56, No. 9, (Sep. 2013), 3 pgs. cited by applicant

Sophia, Bernazzani, "A Brief History of Snapchat", Hubspot, [Online] Retrieved from the Internet: <URL: <https://blog.hubspot.com/marketing/history-of-snapchat>>, (Feb. 10, 2017), 12 pgs. cited by applicant

Sulleyman, Aatif, "Google Maps Could Let Strangers Track Your Real-Time Location For Days at a Time", The Independent, [Online] Retrieved from the Internet: <URL:

<https://www.independent.co.uk/life-style/gadgets-and-tech/news/google-maps-track-location-real-time-days-privacy-security-stalk-gps-days-a7645721.html>>, (Mar. 23, 2017), 5 pgs. cited by applicant

Tripathi, Rohit, "Watermark Images in PHP And Save File on Server", [Online] Retrieved from the

Internet: <URL: <http://code.rohitink.com/2012/12/28/watermark-images-in-php-and-save-file-on-server>>, (Dec. 28, 2012), 4 pgs. cited by applicant

Vaas, Lisa, "StealthText, Should You Choose to Accept It", [Online] Retrieved from the Internet: <URL: <http://www.eweek.com/print/c/a/MessagingandCollaboration/StealthTextShouldYouChoosetoAcceptIt>>, (Dec. 13, 2005), 2 pgs. cited by applicant

Zibreg, "How to share your real time location on Google Maps", idownloadblog.com, [Online] Retrieved from the Internet: <URL: <https://www.idownloadblog.com/2017/04/12/how-to-share-location-google-maps/>>, [Retrieved on: Oct. 23, 2019], (Apr. 12, 2017), 23 pgs. cited by applicant

U.S. Appl. No. 17/314,963, filed May 7, 2021, Generating and Displaying Customized Avatars in Media Overlays. cited by applicant

"U.S. Appl. No. 12/471,811, Advisory Action mailed Mar. 28, 2012", 6 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Examiner Interview Summary mailed Feb. 2, 2012", 3 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Examiner Interview Summary mailed Apr. 18, 2011", 3 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Examiner Interview Summary mailed May 27, 2014", 2 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Final Office Action mailed Dec. 23, 2011", 20 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Non Final Office Action mailed Jan. 13, 2011", 15 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Non Final Office Action mailed Jun. 28, 2011", 26 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Non Final Office Action mailed Oct. 24, 2014", 21 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Notice of Allowance mailed Apr. 1, 2015", 6 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Response filed Jan. 26, 2015 to Non Final Office Action mailed Oct. 24, 2014", 18 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Response filed Feb. 23, 2012 to Final Office Action mailed Dec. 23, 2011", 12 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Response filed Mar. 28, 2012 to Advisory Action mailed Mar. 28, 2012", 14 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Response filed Apr. 13, 2011 to Non Final Office Action mailed Jan. 13, 2011", 5 pgs. cited by applicant

"U.S. Appl. No. 12/471,811, Response filed Sep. 28, 2011 to Non Final Office Action mailed Jun. 28, 2011", 19 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Corrected Notice of Allowability mailed Nov. 19, 2018", 2 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Examiner Interview Summary mailed Jun. 29, 2017", 3 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Examiner Interview Summary mailed Sep. 15, 2017", 3 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Final Office Action mailed Apr. 25, 2018", 18 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Final Office Action mailed Jun. 9, 2017", 20 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Final Office Action mailed Oct. 12, 2016", 13 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Non Final Office Action mailed Feb. 22, 2017", 17 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Non Final Office Action mailed Apr. 27, 2016", 16 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Non Final Office Action mailed Oct. 3, 2017", 17 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Notice of Allowance mailed Aug. 10, 2018", 9 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Response filed Jan. 3, 2018 to Non Final Office Action mailed Oct. 3, 2017", 8 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Response filed May 22, 2017 to Non Final Office Action mailed Feb. 22, 2017", 10 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Response filed Jul. 25, 2018 to Final Office Action mailed Apr. 25, 2018", 10 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Response filed Jul. 26, 2016 to Non Final Office Action mailed Apr. 27, 2016", 8 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Response filed Sep. 11, 2017 to Final Office Action mailed Jun. 9, 2017", 8 pgs. cited by applicant

"U.S. Appl. No. 13/979,974, Response filed Jan. 12, 2017 to Non Final Office Action mailed Apr. 27, 2016", 8 pgs. cited by applicant

"U.S. Appl. No. 14/753,200, Non Final Office Action mailed Oct. 11, 2016", 6 pgs. cited by applicant

"U.S. Appl. No. 14/753,200, Notice of Allowance mailed Apr. 27, 2017", 7 pgs. cited by applicant

"U.S. Appl. No. 14/753,200, Response filed Feb. 13, 2017 to Non Final Office Action mailed Oct. 11, 2016", 9 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Final Office Action mailed Oct. 31, 2017", 15 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Final Office Action mailed Dec. 31, 2018", 14 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Non Final Office Action mailed Mar. 13, 2017", 12 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Non Final Office Action mailed Apr. 30, 2018", 14 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Notice of Allowance mailed Feb. 26, 2019", 7 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Response filed Feb. 11, 2019 to Final Office Action mailed Dec. 31, 2018", 10 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Response filed Apr. 2, 2018 to Final Office Action mailed Oct. 31, 2017", 14 pgs. cited by applicant

"U.S. Appl. No. 15/086,749, Response filed Aug. 29, 2018 to Non Final Office Action mailed Apr. 30, 2018", 12 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Final Office Action mailed Mar. 1, 2018", 31 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Final Office Action mailed Nov. 15, 2018", 37 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Non Final Office Action mailed Jul. 25, 2017", 30 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Non Final Office Action mailed Sep. 21, 2018", 33 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Notice of Allowability mailed May 13, 2019", 3 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Notice of Allowance mailed Mar. 18, 2019", 23 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Response filed Jan. 15, 2019 to Final Office Action mailed Nov. 15, 2018", 14 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Response filed Jan. 25, 2018 to Non Final Office Action mailed Jul. 25, 2017", 13 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Response filed Aug. 31, 2018 to Final Office Action mailed Mar. 1, 2018", 14 pgs. cited by applicant

"U.S. Appl. No. 15/199,472, Response filed Oct. 17, 2018 to Non Final Office Action mailed Sep. 31, 2018", 11 pgs. cited by applicant

"U.S. Appl. No. 15/365,046, Non Final Office Action mailed Dec. 20, 2018", 36 pgs. cited by applicant

"U.S. Appl. No. 15/365,046, Notice of Allowance mailed May 21, 2019", 14 pgs. cited by applicant

"U.S. Appl. No. 15/365,046, Response filed Mar. 20, 2019 to Non Final Office Action mailed Dec. 20, 2018", 20 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Examiner Interview Summary mailed Sep. 21, 2020", 3 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Examiner Interview Summary mailed Oct. 9, 2020", 2 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Final Office Action mailed Jan. 31, 2019", 22 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Final Office Action mailed Jun. 15, 2020", 17 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Final Office Action mailed Oct. 1, 2019", 17 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Non Final Office Action mailed Mar. 2, 2020", 17 pgs. cited by applicant

"U.S. Appl. No. 15/369,499, Non Final Office Action mailed Jun. 17, 2019", 17 pgs. cited by applicant

“U.S. Appl. No. 15/369,499, Non Final Office Action mailed Aug. 15, 2018”, 22 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Notice of Allowance mailed Oct. 26, 2020”, 17 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Response filed Feb. 3, 2020 to Final Office Action mailed Oct. 1, 2019”, 10 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Response filed Mar. 14, 2019 to Final Office Action mailed Jan. 31, 2019”, 12 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Response filed Jun. 2, 2020 to Non Final Office Action mailed Mar. 2, 2020”, 9 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Response filed Sep. 15, 2020 to Final Office Action mailed Jun. 15, 2020”, 10 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Response filed Nov. 15, 2018 to Non Final Office Action mailed Aug. 15, 2018”, 10 pgs. cited by applicant
“U.S. Appl. No. 15/369,499, Response filed Sep. 10, 2019 to Non-Final Office Action mailed Jun. 17, 2019”, 9 pgs. cited by applicant
“U.S. Appl. No. 15/401,926, Response filed May 20, 2019 to Restriction Requirement mailed Mar. 29, 2019”, 9 pgs. cited by applicant
“U.S. Appl. No. 15/401,926, Restriction Requirement mailed Mar. 29, 2019”, 7 pgs. cited by applicant
“U.S. Appl. No. 15/583,142, Jan. 28, 2019 to Response Filed Non Final Office Action mailed Oct. 25, 2018”, 19 pgs. cited by applicant
“U.S. Appl. No. 15/583,142, Final Office Action mailed Mar. 22, 2019”, 11 pgs. cited by applicant
“U.S. Appl. No. 15/583,142, Non Final Office Action mailed Oct. 25, 2018”, 14 pgs. cited by applicant
“U.S. Appl. No. 15/583,142, Response filed May 9, 2019 to Final Office Action mailed Mar. 22, 2019”, 8 pgs. cited by applicant
“U.S. Appl. No. 15/628,408, Non Final Office Action mailed Jan. 2, 2019”, 28 pgs. cited by applicant
“U.S. Appl. No. 15/628,408, Response filed Apr. 2, 2019 to Non Final Office Action mailed Jan. 2, 2019”, 15 pgs. cited by applicant
“U.S. Appl. No. 15/628,408, Supplemental Amendment filed Apr. 4, 2019 to Non Final Office Action mailed Jan. 2, 2019”, 12 pgs. cited by applicant
“U.S. Appl. No. 15/661,953, Examiner Interview Summary mailed Nov. 13, 2018”, 3 pgs. cited by applicant
“U.S. Appl. No. 15/661,953, Non Final Office Action mailed Mar. 26, 2018”, 6 pgs. cited by applicant
“U.S. Appl. No. 15/661,953, Notice of Allowance mailed Aug. 10, 2018”, 7 pgs. cited by applicant
“U.S. Appl. No. 15/661,953, PTO Response to Rule 312 Communication mailed Oct. 30, 2018”, 2 pgs. cited by applicant
“U.S. Appl. No. 15/661,953, PTO Response to Rule 312 Communication mailed Nov. 7, 2018”, 2 pgs. cited by applicant
“U.S. Appl. No. 15/661,953, Response Filed Jun. 26, 2018 to Non Final Office Action mailed Mar. 26, 2018”, 13 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Final Office Action mailed Jul. 22, 2020”, 20 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Final Office Action mailed Dec. 16, 2019”, 23 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Non Final Office Action mailed Apr. 9, 2020”, 18 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Non Final Office Action mailed Jul. 30, 2019”, 21 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Preliminary Amendment filed Oct. 18, 2018 t”, 6 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Response filed Mar. 13, 2020 to Final Office Action mailed Dec. 16, 2019”, 9 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Response filed Jul. 9, 2020 to Non Final Office Action mailed Apr. 9, 2020”, 8 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Response filed Oct. 22, 2020 to Final Office Action mailed Jul. 22, 2020”, 10 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Response filed Oct. 30, 2019 to Non Final Office Action mailed Jul. 30,

2019", 9 pgs. cited by applicant

"U.S. Appl. No. 16/193,938, Preliminary Amendment filed Nov. 27, 2018", 7 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Examiner Interview Summary mailed Jul. 20, 2020", 4 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Final Office Action mailed Jun. 2, 2020", 29 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Non Final Office Action mailed Feb. 27, 2020", 34 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Non Final Office Action mailed Aug. 20, 2020", 29 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Notice of Allowance mailed Dec. 16, 2020", 8 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Response filed May 8, 2020 to Non Final Office Action mailed Feb. 27, 2020", 13 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Response filed Aug. 3, 2020 to Final Office Action mailed Jun. 2, 2020", 12 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Response filed Nov. 16, 2020 to Non Final Office Action mailed Aug. 20, 2020", 12 pgs. cited by applicant

"U.S. Appl. No. 16/433,725, Supplemental Notice of Allowability mailed Jan. 25, 2021", 2 pgs. cited by applicant

"European Application Serial No. 17751497.3, Response filed May 20, 2019 to Communication pursuant to Rules 161(1) and 162 EPC mailed Feb. 14, 2019", w/ English Claims, 24 pgs. cited by applicant

"European Application Serial No. 17776809.0, Extended European Search Report mailed Feb. 27, 2019", 7 pgs. cited by applicant

"International Application Serial No. PCT/CA2013/000454, International Preliminary Report on Patentability mailed Nov. 20, 2014", 9 pgs. cited by applicant

"International Application Serial No. PCT/CA2013/000454, International Search Report mailed Aug. 20, 2013", 3 pgs. cited by applicant

"International Application Serial No. PCT/CA2013/000454, Written Opinion mailed Aug. 20, 2013", 7 pgs. cited by applicant

"International Application Serial No. PCT/US2017/025460, International Preliminary Report on Patentability mailed Oct. 11, 2018", 9 pgs. cited by applicant

"International Application Serial No. PCT/US2017/025460, International Search Report mailed Jun. 20, 2017", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2017/025460, Written Opinion mailed Jun. 20, 2017", 7 pgs. cited by applicant

"International Application Serial No. PCT/US2017/040447, International Preliminary Report on Patentability mailed Jan. 10, 2019", 8 pgs. cited by applicant

"International Application Serial No. PCT/US2017/040447, International Search Report mailed Oct. 2, 2017", 4 pgs. cited by applicant

"International Application Serial No. PCT/US2017/040447, Written Opinion mailed Oct. 2, 2017", 6 pgs. cited by applicant

"International Application Serial No. PCT/US2017/057918, International Preliminary Report on Patentability mailed May 9, 2019", 9 pgs. cited by applicant

"International Application Serial No. PCT/US2017/057918, International Search Report mailed Jan. 19, 2018", 3 pgs. cited by applicant

"International Application Serial No. PCT/US2017/057918, Written Opinion mailed Jan. 19, 2018", 7 pgs. cited by applicant

"International Application Serial No. PCT/US2017/063981, International Search Report mailed Mar. 22, 2018", 3 pgs. cited by applicant

"International Application Serial No. PCT/US2017/063981, Written Opinion mailed Mar. 22, 2018", 8 pgs. cited by applicant

"International Application Serial No. PCT/US2018/000112, International Search Report mailed Jul. 20,

2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/000112, Written Opinion mailed Jul. 20, 2018", 4 pgs. cited by applicant

"International Application Serial No. PCT/US2018/000113, International Search Report mailed Jul. 13, 2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/000113, Written Opinion mailed Jul. 13, 2018", 4 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030039, International Search Report mailed Jul. 11, 2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030039, Written Opinion mailed Jul. 11, 2018", 4 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030043, International Search Report mailed Jul. 23, 2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030043, Written Opinion mailed Jul. 23, 2018", 5 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030044, International Search Report mailed Jun. 26, 2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030044, Written Opinion mailed Jun. 26, 2018", 6 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030045, International Search Report mailed Jul. 3, 2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030045, Written Opinion mailed Jul. 3, 2018", 6 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030046, International Search Report mailed Jul. 6, 2018", 2 pgs. cited by applicant

"International Application Serial No. PCT/US2018/030046, Written Opinion mailed Jul. 6, 2018", 6 pgs. cited by applicant

"Korean Application Serial No. 10-2019-7002736, Final Office Action mailed Nov. 26, 2020", w/ English Translation, 8 pgs. cited by applicant

"Korean Application Serial No. 10-2019-7002736, Notice of Preliminary Rejection mailed May 25, 2020", W/English Translation, 16 pgs. cited by applicant

"Korean Application Serial No. 10-2019-7002736, Response filed Jul. 9, 2020 to Notice of Preliminary Rejection mailed May 25, 2020", w/ English Claims, 29 pgs. cited by applicant

"List of IBM Patents or Patent Applications Treated as Related; {Appendix P}", IBM, (Sep. 14, 2018), 2 pgs. cited by applicant

Broderick, Ryan, "Every thing You Need to Know About Japan's Amazing Photo Booths", [Online] Retrieved from the Internet: <URL: https://www.buzzfeed.com/ryanhatesthis/look-how-kawaii-i-am?utm_term=.kra5QwGNZ#.muYoVB7qJ>, (Jan. 22, 2016), 30 pgs. cited by applicant

Chan, Connie, "The Elements of Stickers", [Online] Retrieved from the Internet: <URL: <https://a16z.com/2016/06/17/stickers/>>, (Jun. 20, 2016), 15 pgs. cited by applicant

Collet, Jean Luc, et al., "Interactive avatar in messaging environment", U.S. Appl. No. 12/471,811, filed May 26, 2009, (filed May 26, 2009), 31 pgs. cited by applicant

Dillet, Romain, "Zenly proves that location sharing isn't dead", [Online] Retrieved from the Internet: <URL: <https://techcrunch.com/2016/05/19/zenly-solomoyolo/>>, (accessed Jun. 27, 2018), 6 pgs. cited by applicant

Leyden, John, "This SMS will self-destruct in 40 seconds", [Online] Retrieved from the Internet: <URL: <http://www.theregister.co.uk/2005/12/12/stealthtext/>>, (Dec. 12, 2005), 1 pg. cited by applicant

Petovello, Mark, "How does a GNSS receiver estimate velocity?", InsideGNSS, [Online] Retrieved from the internet: <<http://insidegnss.com/wp-content/uploads/2018/01/marapr15-SOLUTIONS.pdf>>., (Mar.- Apr. 2015), 3 pgs. cited by applicant

Rhee, Chi-Hyoung, et al., "Cartoon-like Avatar Generation Using Facial Component Matching", International Journal of Multimedia and Ubiquitous Engineering, (Jul. 30, 2013), 69-78. cited by applicant

"U.S. Appl. No. 16/115,259, Final Office Action mailed Apr. 4, 2022", 18 pgs. cited by applicant

"U.S. Appl. No. 16/115,259, Non Final Office Action mailed Nov. 8, 2021", 17 pgs. cited by applicant

"U.S. Appl. No. 16/115,259, Response filed Feb. 8, 2022 to Non Final Office Action mailed Nov. 8, 2021", 9 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Non Final Office Action mailed Feb. 2, 2022", 24 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Response filed May 2, 2022 to Non Final Office Action mailed Feb. 2, 2022", 10 pgs. cited by applicant

"European Application Serial No. 17751497.3, Response filed Sep. 16, 2021 to Communication Pursuant to Article 94(3) EPC mailed Jun. 2, 2021", 15 pgs. cited by applicant

"Korean Application Serial No. 10-2021-7010821, Final Office Action mailed Nov. 29, 2021", w/ English translation, 7 pgs. cited by applicant

"Korean Application Serial No. 10-2021-7014438, Notice of Preliminary Rejection mailed Aug. 9, 2021", w/ English Translation, 4 pgs. cited by applicant

"Korean Application Serial No. 10-2021-7014438, Response filed Oct. 12, 2021 to Notice of Preliminary Rejection mailed Aug. 9, 2021", w/ English Claims, 15 pgs. cited by applicant

"Korean Application Serial No. 10-2021-7010821, Response filed Dec. 30, 2021 to Office Action mailed Nov. 29, 2021", w/ English Translation of Claims, 22 pgs. cited by applicant

"U.S. Appl. No. 16/115,259, Response filed Sep. 6, 2022 to Final Office Action mailed Apr. 4, 2022", 10 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Final Office Action mailed Jul. 11, 2022", 25 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Response filed Sep. 12, 2022 to Final Office Action mailed Jul. 11, 2022", 11 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Advisory Action mailed Sep. 27, 2022", 3 pgs. cited by applicant

"Korean Application Serial No. 10-2022-7015893, Notice of Preliminary Rejection mailed Aug. 11, 2022", W/English Translation, 6 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Response filed Oct. 11, 2022 to Advisory Action mailed Sep. 27, 2022", 10 pgs. cited by applicant

"U.S. Appl. No. 16/115,259, Non Final Office Action mailed Nov. 1, 2022", 18 pgs. cited by applicant

"Korean Application Serial No. 10-2022-7015893, Response filed Oct. 7, 2022 to Notice of Preliminary Rejection mailed Aug. 11, 2022", w English Claims, 17 pgs. cited by applicant

"European Application Serial No. 17751497.3, Summons to Attend Oral Proceedings mailed Dec. 20, 2022", 5 pgs. cited by applicant

"Chinese Application Serial No. 201780065441.1, Office Action mailed Dec. 2, 2022", w/ English Translation, 13 pgs. cited by applicant

"Chinese Application Serial No. 201780052571.1, Office Action mailed Dec. 5, 2022", w/ English Translation, 15 pgs. cited by applicant

"Chinese Application Serial No. 201780065441.1, Office Action mailed Dec. 5, 2022", w/ English Translation, 13 pgs. cited by applicant

Vaynerchuk, Gary, "How to Create and Use Snapchat's New Custom Geofilters", [Online] Retrieved from the Internet: <<https://garyvaynerchuk.com/how-to-create-and-use-snapchats-new-custom-geofilters/>>, (Mar. 15, 2016), 15 pgs. cited by applicant

"U.S. Appl. No. 16/115,259, Examiner Interview Summary mailed Feb. 16, 2023", 2 pgs. cited by applicant

"U.S. Appl. No. 16/115,259, Response filed Feb. 1, 2023 to Non Final Office Action mailed Nov. 1, 2022", 10 pgs. cited by applicant

"U.S. Appl. No. 17/314,963, Corrected Notice of Allowability mailed Jan. 26, 2023", 2 pgs. cited by applicant

“U.S. Appl. No. 17/314,963, Notice of Allowance mailed Jan. 13, 2023”, 6 pgs. cited by applicant
“Application Serial No. 4218.394US2, Notice of Allowance mailed Feb. 27, 2023”, 8 pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Corrected Notice of Allowability mailed Jun. 12, 2023”, 3 pgs. cited by applicant
“U.S. Appl. No. 17/314,963, Notice of Allowance mailed Apr. 14, 2023”, 5 pgs. cited by applicant
“Chinese Application Serial No. 201780065441.1, Office Action mailed May 26, 2023”, w/ English Translation, 14 pgs. cited by applicant
“U.S. Appl. No. 17/314,963, Notice of Allowance mailed Aug. 2, 2023”, 5 pgs. cited by applicant
“Chinese Application Serial No. 201780052571.1, Office Action mailed Jun. 30, 2023”, W/English Translation, 9 pgs. cited by applicant
“European Application Serial No. 17751497.3, EPO Written Decision to Refuse mailed Jul. 18, 2023”, 11 pgs. cited by applicant
“Korean Application Serial No. 10-2023-7015209, Notice of Preliminary Rejection mailed Jul. 3, 2023”, W/English Translation, 6 pgs. cited by applicant
U.S. Appl. No. 18/138,552, filed Apr. 24, 2023, Generating and Displaying Customized Avatars in Media Overlays. cited by applicant
“Chinese Application Serial No. 201780052571.1, Response Filed Sep. 15, 2023 To Office Action mailed Jun. 30, 2023”, W/English Claims, 12 pgs. cited by applicant
“Chinese Application Serial No. 201780065441.1, Office Action mailed Aug. 18, 2023”, w/ English Translation, 6 pgs. cited by applicant
“European Application Serial No. 23191199.1, Extended European Search Report mailed Nov. 3, 2023”, 6 pgs. cited by applicant
“Korean Application Serial No. 10-2023-7015209, Response Filed Sep. 4, 2023 To Notice of Preliminary Rejection mailed Jul. 3, 2023”, W/ English Claims, 19 Pgs. cited by applicant
“U.S. Appl. No. 16/115,259, Notice of Allowability mailed Dec. 14, 2023”, 3 pgs. cited by applicant
“U.S. Appl. No. 18/138,552, Corrected Notice of Allowability mailed Jan. 4, 2024”, 2 pgs. cited by applicant
“U.S. Appl. No. 18/138,552, Notice of Allowance mailed Dec. 20, 2023”, 6 pgs. cited by applicant
“Chinese Application Serial No. 201780052571.1, Decision of Rejection mailed Nov. 29, 2023”, w/ English Translation, 12 pgs. cited by applicant
“Chinese Application Serial No. 201780065441.1, Response filed Apr. 7, 2023 to Office Action mailed Dec. 2, 2022”, w/ current English claims, 9 pgs. cited by applicant
“Chinese Application Serial No. 201780065441.1, Response filed Jul. 25, 2023 to Office Action mailed May 26, 2023”, w/ English claims, 13 pgs. cited by applicant
“Chinese Application Serial No. 201780065441.1, Response filed Oct. 31, 2023 to Office Action mailed Aug. 18, 2023”, w/ English claims, 13 pgs. cited by applicant
“U.S. Appl. No. 18/138,552, Notice of Allowance mailed Apr. 9, 2024”, 5 pgs. cited by applicant
“Korean Application Serial No. 10-2024-7008682, Notice of Preliminary Rejection mailed May 16, 2024”, W/English Translation, 6 pgs. cited by applicant
“U.S. Appl. No. 18/138,552, Corrected Notice of Allowability mailed Sep. 9, 2024”, 2 pgs. cited by applicant
“U.S. Appl. No. 18/586,080, Non Final Office Action mailed Sep. 18, 2024”, 46 pgs. cited by applicant
“European Application Serial No. 23191199.1, Response filed Jun. 4, 2024 to Extended European Search Report mailed Nov. 3, 2023”, 19 pgs. cited by applicant

Primary Examiner: Lhymn; Sarah

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

PRIORITY CLAIM (1) This application is a continuation of and claims the benefit of priority of U.S. patent application Ser. No. 16/433,725, filed on Jun. 6, 2019, which is a continuation of and claims the benefit of priority of U.S. patent application Ser. No. 15/199,472, filed on Jun. 30, 2016, each of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) Embodiments of the present disclosure relate generally to automated processing of images. More particularly, but not by way of limitation, the present disclosure addresses systems and methods for generating ideogram representations of a face depicted within a set of images.

BACKGROUND

(2) Telecommunications applications and devices can provide communication between multiple users using a variety of media, such as text, images, sound recordings, and/or video recording. For example, video conferencing allows two or more individuals to communicate with each other using a combination of software applications, telecommunications devices, and a telecommunications network. Telecommunications devices may also record video streams to transmit as messages across a telecommunications network.

(3) Currently ideograms in telecommunication applications are centrally generated by entities distributing applications or brands releasing licensed content. Ideograms are provided in telecommunication applications in set packages or individual downloads.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various ones of the appended drawings merely illustrate example embodiments of the present disclosure and should not be considered as limiting its scope.

(2) FIG. 1 is a block diagram illustrating a networked system, according to some example embodiments.

(3) FIG. 2 is a diagram illustrating an ideogram generation system, according to some example embodiments.

(4) FIG. 3 is a flow diagram illustrating an example method for generating an ideogram from a set of images of an image stream, according to some example embodiments.

(5) FIG. 4 is a user interface diagram depicting a face within a video stream and a generated graphical model.

(6) FIG. 5 is a user interface diagram depicting an ideogram generated from a graphical model.

(7) FIG. 6 is a flow diagram illustrating an example method for generating an ideogram from a set of images of an image stream, according to some example embodiments.

(8) FIG. 7 is a flow diagram illustrating an example method for generating an ideogram from a set of images of an image stream, according to some example embodiments.

(9) FIG. 8 is a flow diagram illustrating an example method for generating an ideogram from a set of images of an image stream, according to some example embodiments.

(10) FIG. 9 is a flow diagram illustrating an example method for generating an ideogram from a set of images of an image stream, according to some example embodiments.

(11) FIG. 10 is a flow diagram illustrating an example method for generating an ideogram from a set of images of an image stream, according to some example embodiments.

(12) FIG. 11 is a user interface diagram depicting an example mobile device and mobile operating system interface, according to some example embodiments.

(13) FIG. 12 is a block diagram illustrating an example of a software architecture that may be

installed on a machine, according to some example embodiments.

(14) FIG. 13 is a block diagram presenting a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any of the methodologies discussed herein, according to an example embodiment.

(15) The headings provided herein are merely for convenience and do not necessarily affect the scope or meaning of the terms used.

DETAILED DESCRIPTION

(16) The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products illustrative of embodiments of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the inventive subject matter. It will be evident, however, to those skilled in the art, that embodiments of the inventive subject matter may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

(17) Although methods exist to generate avatars or representations of faces within an image, most of these methods do not employ facial recognition or facial landmarks as a basis for the generated avatar or representation of the face. Although methods exist to generate ideograms for use in telecommunication applications, these methods are not generated from avatars or image streams. Further, these methods do not generate ideograms from image streams captured in real time on a client device. Generation of ideograms is often performed by entities distributing telecommunication applications. The ideograms are then distributed to users via the telecommunications application. These ideograms provide no customization and do not reflect avatars or images associated with a user. Accordingly, there is still a need in the art to improve generation of avatars and ideograms without user interaction or with minimal user interaction. Further, there is still a need in the art to improve generation of stylized (e.g., animated and cartoon image) ideograms which are reasonable facsimiles of a face depicted within an image using facial landmarks derived from the face and measurements generated based on the facial landmarks. As described herein, methods and systems are presented for generating facial avatars or ideograms based on facial landmarks of a face depicted within an image using a user interaction of an initial selection.

(18) Embodiments of the present disclosure may relate generally to automated image segmentation and generation of facial representations within an ideogram based on the segmented image. In one embodiment, a user of a client device may open an application operating on the client device. Selection of a user interface element by the user causes capture of an image using a camera of the client device. The user may then select a “generate sticker” button within the application to cause the application to build an avatar using the captured image and enable generation of an ideogram based on the avatar. The application may identify facial landmarks, measurements between facial landmarks, and characteristics of the face to generate a look-alike avatar based on the image and proportions of the face. After generating the avatar, the application may present buttons enabling the user to save the avatar, manipulate or customize the avatar, and an ideogram. The ideogram may include digital stickers, emojis, animated bitmap images, and other graphics which may be shared with other users by including the graphics in messages or other communications between client devices.

(19) The above is one specific example. The various embodiments of the present disclosure relate to devices and instructions by one or more processors of a device to modify an image or a video stream transmitted by the device to another device while the video stream is being captured (e.g., modifying a video stream in real time). An ideogram generation system is described that identifies and tracks objects and areas of interest within an image or across a video stream and through a set of images comprising the video stream. In various example embodiments, the ideogram generation system identifies and tracks one or more facial features depicted in a video stream or within an

image and performs image recognition, facial recognition, facial processing functions with respect to the one or more facial features and interrelations between two or more facial features, and generation of ideograms from the avatar and the tracked facial features.

(20) FIG. 1 is a network diagram depicting a network system **100** having a client-server architecture configured for exchanging data over a network, according to one embodiment. For example, the network system **100** may be a messaging system where clients communicate and exchange data within the network system **100**. The data may pertain to various functions (e.g., sending and receiving text and media communication, determining geolocation, etc.) and aspects (e.g., transferring communications data, receiving and transmitting indications of communication sessions, etc.) associated with the network system **100** and its users. Although illustrated herein as client-server architecture, other embodiments may include other network architectures, such as peer-to-peer or distributed network environments.

(21) As shown in FIG. 1, the network system **100** includes a social messaging system **130**. The social messaging system **130** is generally based on a three-tiered architecture, consisting of an interface layer **124**, an application logic layer **126**, and a data layer **128**. As is understood by skilled artisans in the relevant computer and Internet-related arts, each component or engine shown in FIG. 1 represents a set of executable software instructions and the corresponding hardware (e.g., memory and processor) for executing the instructions, forming a hardware-implemented component or engine and acting, at the time of the execution of instructions, as a special purpose machine configured to carry out a particular set of functions. To avoid obscuring the inventive subject matter with unnecessary detail, various functional components and engines that are not germane to conveying an understanding of the inventive subject matter have been omitted from FIG. 1. Of course, additional functional components and engines may be used with a social messaging system, such as that illustrated in FIG. 1, to facilitate additional functionality that is not specifically described herein. Furthermore, the various functional components and engines depicted in FIG. 1 may reside on a single server computer or client device, or may be distributed across several server computers or client devices in various arrangements. Moreover, although the social messaging system **130** is depicted in FIG. 1 as a three-tiered architecture, the inventive subject matter is by no means limited to such an architecture.

(22) As shown in FIG. 1, the interface layer **124** consists of interface component(s) (e.g., a web server) **140**, which receives requests from various client-computing devices and servers, such as client device **110** executing client application(s) **112**, and third party server(s) **120** executing third party application(s) **122**. In response to received requests, the interface component(s) **140** communicates appropriate responses to requesting devices via a network **104**. For example, the interface component(s) **140** can receive requests such as Hypertext Transfer Protocol (HTTP) requests, or other web-based, Application Programming Interface (API) requests.

(23) The client device **110** can execute conventional web browser applications or applications (also referred to as “apps”) that have been developed for a specific platform to include any of a wide variety of mobile computing devices and mobile-specific operating systems (e.g., IOS™, ANDROID™, WINDOWS® PHONE). Further, in some example embodiments, the client device **110** forms all or part of an ideogram generation system **160** such that components of the ideogram generation system **160** configure the client device **110** to perform a specific set of functions with respect to operations of the ideogram generation system **160**.

(24) In an example, the client device **110** is executing the client application(s) **112**. The client application(s) **112** can provide functionality to present information to a user **106** and communicate via the network **104** to exchange information with the social messaging system **130**. Further, in some examples, the client device **110** executes functionality of the ideogram generation system **160** to segment images of video streams during capture of the video streams and transmits the video streams (e.g., with image data modified based on the segmented images of the video stream) or generates image representations (e.g., ideograms) from data included in the video stream.

(25) Each client device **110** can comprise a computing device that includes at least a display and communication capabilities with the network **104** to access the social messaging system **130**, other client devices, and third party server(s) **120**. Client devices **110** comprise, but are not limited to, remote devices, work stations, computers, general purpose computers, Internet appliances, hand-held devices, wireless devices, portable devices, wearable computers, cellular or mobile phones, personal digital assistants (PDAs), smart phones, tablets, ultrabooks, netbooks, laptops, desktops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, network PCs, mini-computers, and the like. User **106** can be a person, a machine, or other means of interacting with the client device **110**. In some embodiments, the user **106** interacts with the social messaging system **130** via the client device **110**. The user **106** may not be part of the networked system **100**, but may be associated with the client devices **110**.

(26) As shown in FIG. 1, the data layer **128** has database server(s) **132** that facilitate access to information storage repositories or database(s) **134**. The database(s) **134** are storage devices that store data such as member profile data, social graph data (e.g., relationships between members of the social messaging system **130**), image modification preference data, accessibility data, and other user data.

(27) An individual can register with the social messaging system **130** to become a member of the social messaging system **130**. Once registered, a member can form social network relationships (e.g., friends, followers, or contacts) on the social messaging system **130** and interact with a broad range of applications provided by the social messaging system **130**.

(28) The application logic layer **126** includes various application logic components **150**, which, in conjunction with the interface component(s) **140**, generate various user interfaces with data retrieved from various data sources or data services in the data layer **128**. Individual application logic components **150** may be used to implement the functionality associated with various applications, services, and features of the social messaging system **130**. For instance, a social messaging application can be implemented with at least a portion of the application logic components **150**. The social messaging application provides a messaging mechanism for users of the client devices **110** to send and receive messages that include text and media content such as pictures and video. The client devices **110** may access and view the messages from the social messaging application for a specified period of time (e.g., limited or unlimited). In an example, a particular message is accessible to a message recipient for a predefined duration (e.g., specified by a message sender) that begins when the particular message is first accessed. After the predefined duration elapses, the message is deleted and is no longer accessible to the message recipient. Of course, other applications and services may be separately embodied in their own application logic components **150**.

(29) As illustrated in FIG. 1, the social messaging system **130** may include at least a portion of the ideogram generation system **160** capable of identifying, tracking, and modifying video data during capture of the video data by the client device **110**. Similarly, the client device **110** includes at least a portion of the ideogram generation system **160**, as described above. In other examples, client device **110** may include the entirety of ideogram generation system **160**. In instances where the client device **110** includes a portion of (or all of) the ideogram generation system **160**, the client device **110** can work alone or in cooperation with the social messaging system **130** to provide the functionality of the ideogram generation system **160** described herein.

(30) In some embodiments, the social messaging system **130** may be an ephemeral message system that enables ephemeral communications where content (e.g., video clips or images) are deleted following a deletion trigger event such as a viewing time or viewing completion. In such embodiments, a device uses the various components described herein within the context of any of generating, sending, receiving, or displaying aspects of an ephemeral message. For example, a device implementing the ideogram generation system **160** may identify, track, and modify an object of interest, such as pixels representing skin on a face depicted in the video clip. The device may

modify the object of interest during capture of the video clip without image processing after capture of the video clip as a part of a generation of content for an ephemeral message.

(31) In FIG. 2, in various embodiments, the ideogram generation system **160** can be implemented as a standalone system or implemented in conjunction with the client device **110**, and is not necessarily included in the social messaging system **130**. The ideogram generation system **160** is shown to include an access component **210**, an identification component **220**, a facial processing component **230**, a characteristic component **240**, an avatar component **250**, and an ideogram component **260**. All, or some, of the components **210-260**, communicate with each other, for example, via a network coupling, shared memory, and the like. Each component of components **210-260** can be implemented as a single component, combined into other components, or further subdivided into multiple components. Other components not pertinent to example embodiments can also be included, but are not shown.

(32) The access component **210** accesses or otherwise retrieves images captured by an image capture device or otherwise received by or stored in the client device **110**. In some instances, the access component **210** may include portions or all of an image capture component configured to cause an image capture device of the client device **110** to capture images based on user interaction with a user interface presented on a display device of the client device **110**. The access component **210** may pass images or portions of images to one or more other components of the ideogram generation system **160**.

(33) The identification component **220** identifies faces or other areas of interest within the image or set of images received from the access component **210**. In some embodiments, the identification component **220** tracks the identified face or areas of interest across multiple images of a set of images (e.g., a video stream). The identification component **220** may pass values (e.g., coordinates within the image or portions of the image) representing the face or areas of interest to one or more components of the ideogram generation system **160**.

(34) The facial processing component **230** identifies facial landmarks depicted on the face or within the areas of interest identified by the identification component **220**. In some embodiments, the facial processing component **230** identifies expected but missing facial landmarks in addition to the facial landmarks which are depicted on the face or within the area of interest. The facial processing component **230** may determine an orientation of the face based on the facial landmarks and may identify one or more relationships between the facial landmarks. The facial processing component **230** may pass values representing the facial landmarks to one or more components of the ideogram generation system **160**.

(35) The characteristic component **240** identifies, determines, or measures one or more characteristics of the face within the image or areas of interest based at least in part on the facial landmarks identified by the facial processing component **230**. In some embodiments, the characteristic component **240** identifies facial features based on the facial landmarks. The characteristic component **240** may determine measurements of the identified facial features and distances extending between two or more facial features. In some embodiments, the characteristic component **240** identifies areas of interest and extracts prevailing colors from the areas of interest identified on the face. The characteristic component **240** may pass values representing the one or more characteristics to the avatar component **250**.

(36) The avatar component **250** generates an avatar or facial representation based on the one or more characteristics received from the characteristic component **240**. In some embodiments, the avatar component **250** generates a stylized representation of the face, such as a cartoon version of the face depicted within the image. The stylized representation may be generated such that the proportions, positions, and prevailing colors of the features identified within the face are matched to the stylized representation. In some embodiments, in order to match the proportions, positions, and prevailing colors, the avatar component **250** independently generates facial feature representations or modifies existing template representations to match the characteristics and facial

features identified by the characteristic component **240**. The avatar component **250** may cause presentation of the finished avatar of a facial representation at a display device of the client device **110**. In some embodiments, the avatar component **250** enables generation of graphics using the generated avatar or facial representation such as stickers, emojis, gifs, and other suitable graphics configured for transmission within a message (e.g., text, short message system messages, instant messages, and temporary messages) to a subsequent client device associated with a subsequent user.

(37) The ideogram component **260** positions graphical elements and a graphical model to generate an ideogram. In some embodiments, the ideogram component **260** positions one or more graphical elements and the graphical model with respect to one another. The ideogram component **260** may also resize one or more of the one or more graphical elements and the graphical model. The ideogram component **260** may resize graphical elements and the graphical model to fit within dimensions of ideograms for a target application.

(38) FIG. 3 depicts a flow diagram illustrating an example method **300** for generating ideograms from a set of images received in an image stream. The operations of method **300** may be performed by components of the ideogram generation system **160**, and are so described below for purposes of illustration.

(39) In operation **310**, the access component **210** receives or otherwise accesses one or more images depicting at least a portion of a face. In some embodiments, the access component **210** receives the one or more images as a video stream captured by an image capture device associated with the client device **110** and presented on a user interface of an avatar generation application. The access component **210** may include the image capture device as a portion of hardware comprising the access component **210**. In these embodiments, the access component **210** directly receives the one or more images or the video stream captured by the image capture device. In some instances, the access component **210** passes all or a part of the one or more images or the video stream (e.g., a set of images comprising the video stream) to one or more components of the ideogram generation system **160**, as described below in more detail.

(40) In operation **320**, the identification component **220** detects the portion of the face depicted within the one or more images. In some embodiments, the identification component **220** includes a set of face tracking operations to identify a face or a portion of a face within the one or more images. The identification component **220** may use the Viola-Jones object detection framework, Eigen-face technique, a genetic algorithm for face detection, edge detection methods, or any other suitable object-class detection method or set of operations to identify the face or portion of the face within the one or more images. Where the one or more images are a plurality of images (e.g., a set of images in a video stream) the face tracking operations of the identification component **220**, after identifying the face or portion of the face in an initial image, may identify changes in position of the face across multiple images of the plurality of images, thereby tracking movement of the face within the plurality of images. Although specific techniques are described, it should be understood that the identification component **220** may use any suitable technique or set of operations to identify the face or portion of the face within the one or more images without departing from the scope of the present disclosure.

(41) In operation **330**, the facial processing component **230** identifies a set of facial landmarks within the portion of the face depicted within the one or more images. In some embodiments, the facial processing component **230** identifies the set of facial landmarks within the portion of the face in a subset of the one or more images. For example, the facial processing component **230** may identify the set of facial landmarks in a set of images (e.g., a first set of images) of a plurality of images, where the portion of the face or the facial landmarks appear in the set of images but not in the remaining images of the plurality of images (e.g., a second set of images). In some embodiments, identification of the facial landmarks may be performed as a sub-operation or part of identification of the face or portion of the face using face tracking operations incorporating the

detection operations described above.

(42) In operation **340**, the characteristic component **240** determines one or more characteristics representing the portion of the face depicted in the one or more images. In some embodiments, the operation **340** is performed in response to detecting the portion of the face, in the operation **320**, and the set of facial landmarks, in the operation **330**. Characteristics representing the portion of the face may include presence or absence of one or more features (e.g., an eye, an eyebrow, a nose, a mouth, and a perimeter of a face) depicted on the portion of the face, relative positions of the one or more features (e.g., positions of features relative to one another or relative to an outline of the portion of the face), measuring portions of the one or more features, and measuring distances between the two or more of the features. In some instances, characteristics of the portion of the face include color of the one or more features depicted on the face, relative color between an area of the portion of the face and one or more features depicted on the portion of the face, presence or absence of an obstruction, presence or absence of hair, presence or absence of a shadow, or any other suitable characteristics of the portion of the face.

(43) In operation **350**, the avatar component **250** generates a graphical model of a face for the at least one portion of the face depicted in the one or more images. In some embodiments, the operation **350** is performed based on (e.g., in response to) the one or more characteristics being determined in the operation **340** and the set of facial landmarks being identified in the operation **330**. Where the characteristics include one or more measurements for the one or more features depicted on the portion of the face, the avatar component **250** may generate the graphical model of the face by rendering a base face and head shape according to the characteristics and the one or more measurements. As shown in FIG. 4, the avatar component **250** may then generate the one or more features depicted on the face **410** and apply the one or more generated features to the base face and head shape to generate the graphical model **420**. Each of the one or more features may be generated to match one or more measurements associated with the specified feature. As shown in FIG. 4, once generated, one or more of the face **410** and the graphical model **420** may be presented or otherwise displayed on the client device **110**.

(44) In operation **360**, the ideogram component **260** positions one or more graphical elements proximate to the graphical model of the face. The one or more graphical elements may be images, filters, animations (e.g., animated graphics or images), symbols, words, or scenes. The one or more graphical elements may be selected from a set of graphical elements. In some instances, the one or more graphical elements are selected by the ideogram component **260**. In some embodiments, the ideogram component **260** receives selection of user interface elements representing the one or more graphical elements. Selection of the user interface elements may cause the ideogram component **260** to retrieve the one or more graphical elements from a database containing the set of graphical elements.

(45) Where the one or more graphical elements are selected by the ideogram component **260**, the ideogram component **260** may select the one or more graphical elements based on an interaction received at the client device **110**. For example, the access component **210** may receive a selection of a user interface element. The user interface element may be an icon, an entry in a list, or other representation of the one or more graphical elements. In some embodiments, the user interface element represents a theme or predefined group of graphical elements. For example, the user interface element may represent a “Happy Birthday” ideogram. The “Happy Birthday” ideogram may include a first graphical element of balloons and a second graphical element with lettering spelling out “Happy Birthday.” Upon receiving selection of the user interface element for the “Happy Birthday” ideogram, the ideogram component **260** may select the first graphical element and the second graphical element from the set of graphical elements stored on a database. The ideogram component **260** may then position the first graphical element and the second graphical element proximate to the graphical model.

(46) Where the ideogram component **260** receives selection of user interface elements of the one or

more graphical element, the ideogram component **260** may initially cause presentation of a set of graphical elements. The ideogram component **260** may receive selections of the one or more graphical elements included in the set of graphical elements. For example, a user of the client device **110** may be presented with the set of graphical elements in a grid or other ordered presentation at a display device of the client device **110**. The user may tap, touch, click, or otherwise select the one or more graphical elements, causing the client device **110** to pass an indication of the selection to the ideogram component **260**. In some embodiments, the ideogram component **260** may position the one or more graphical elements proximate to the graphical model based on position data of the one or more graphical elements.

(47) In some instances, the ideogram component **260** may receive a position selection indicating placement of the one or more graphical elements with respect to the graphical model. For example, the user may drag the one or more graphical elements to positions proximate to the graphical model using a mouse, keyboard commands, or a touch screen. The positions selected by the user may be predetermined optional positions or may be freely selected by the user. By way of example, upon selection of the one or more graphical elements, the ideogram component **260** may generate instructions for available positions, among the predetermined optional positions, for each of the one or more graphical elements. The instructions may be text instructions, one or more outlines of a graphical element proximate to the graphical model, or any other suitable instruction or indication of a position at which a graphical element may be placed. The user may position the one or more graphical elements, based on the instructions, using a display device and a user input component (e.g., keyboard, mouse, or touch screen). Positioning of the one or more graphical elements causes the client device **110** to pass the positions or data representing the positions to the ideogram component **260**, and the ideogram component **260** may apply or temporarily store the selected positions.

(48) In operation **370**, the ideogram component **260** generates an ideogram from the graphical model and the one or more graphical elements. As shown in FIG. 5, the ideogram **500** may be generated as including the graphical model **420** and the one or more graphical elements **510** and **520**. The ideogram **500** may be generated as a digital sticker, an emoji, an image, or any other suitable ideogram. The ideogram component **260** may generate the ideogram **500** by combining the graphical model **420** and the one or more graphical elements **510**, **520** into a single layered or unlayered ideogram **500**. The ideogram **500** may be generated by inserting the graphical model **420** into a template graphic including the one or more graphical elements **510**, **520**. In these instances, the graphical model **420** may be inserted into a predetermined position with respect to the one or more graphical elements **510**, **520**. In some instances, the ideogram **500** may be animated such that one or more of the one or more graphical elements **510**, **520** and the graphical model **500** move with respect to another of the graphical elements **510**, **520** or the graphical model **420**. For example, the ideogram **500** may be generated such that a first graphical element (e.g., **510**) and the graphical model **420** are animated (e.g., move between one or more predetermined positions) with respect to a second graphical element (e.g., **520**). In some embodiments, animated ideograms may be generated using a set of graphical models in a stream of individual graphical model poses or positions.

(49) In some instances, the ideogram component **260** generates the ideogram (e.g., ideogram **500**) irrespective of dimensions or configuration information of any specific program, application, or set of instructions outside of the ideogram generation system **160**. For example, the ideogram component **260** may generate the ideogram with dimensions (e.g., height and width dimensions, pixel dimensions, or total pixel count) suitable for the ideogram generation system **160** without regard to another application which may use or receive the ideogram. In some instances, the ideogram may be generated using universal configuration information suitable for use across a set of applications (e.g., web browsers, messaging applications, social networking applications, or ephemeral messaging applications). As will be explained below in more detail, the ideogram

component **260** may generate the ideogram based on configuration information of a specified destination application. For example, the ideogram may be generated with dimensions and formatting compliant with a specified messaging or social networking application selected by a user or predetermined at initiation of the ideogram generation system **160**.

(50) In some example embodiments, as part of operation **370**, the ideogram component **260** determines one or more sizes of the one or more graphical elements (e.g., graphical elements **510**, **520**). The ideogram component **260** then scales the graphical model (e.g., graphical model **420**) to generate a scaled graphical model based on the one or more sizes of the one or more graphical elements. In some embodiments, the ideogram component **260** may identify a maximum size of the ideogram and scale one or more of the graphical elements and the graphical model to fit within the maximum size, such that the one or more graphical elements and the graphical model maintain the same or similar relative proportions before and after the scaling. The ideogram component **260** may scale the graphical model and the one or more graphical elements by subsampling or downsampling the graphical model or the one or more graphical elements being scaled. Although described as using downsampling, it should be understood that the ideogram component **260** may use any suitable digital image scaling process, technique, algorithm, or operations suitable to reduce the size of one or more of the graphical model and the one or more graphical elements.

(51) In some embodiments, the ideogram component **260** generates the ideogram by performing a set of ideogram generation operations to render the ideogram from the graphical model and the one or more graphical elements. The ideogram component **260** may first generate an alpha mask. In generating the alpha mask, the ideogram component **260** renders a mesh for the graphical model in a first color on a background having a second color. The first color and the second color may be selected based on a contrast value between the first color and the second color. For example, the first color may be white and the second color may be black. The alpha mask may represent the graphical model bounded within an outline of the graphical model, such that generation of the alpha mask may be a silhouette of the graphical model colored in the first color and positioned on a background of the second color.

(52) In response to generating the alpha mask, the ideogram component **260** generates a graphical model texture. In generating the graphical model texture, the ideogram component **260** renders the graphical model mesh using one or more shading operations. The shading operations may include skin shading, eye shading, hair shading, and other shading operations. In some embodiments, the one or more shading operations are Open Graphics Library (OpenGL) shading operations or are compatible with usage of OpenGL sample coverage features.

(53) After generating the graphical model texture, the ideogram component **260** generates the ideogram from the graphical model, including the generated graphical model texture, the alpha mask, and the one or more graphical elements. In some embodiments, the ideogram component **260** renders the ideogram with a sticker shader function. The sticker shader function may receive texture inputs for layers. In some instances the sticker shader receives texture inputs including the graphical model texture, the alpha mask, and the one or more graphical elements.

(54) In some embodiments, the sticker shader receives texture inputs including the graphical model texture, the alpha mask, and one or more elements for ideogram layers. The elements for the ideogram layers may include a sticker mask layer, a sticker background layer, and a sticker foreground layer. The sticker mask layer, the sticker background layer, and the sticker foreground layer may be variable layers which may or may not be included in a generated ideogram. The variable sticker layers may be included in the generated ideogram where a graphical element corresponds to the sticker layer to be included.

(55) In some embodiments, the ideogram component **260**, in performing the sticker shader function, determines red, green, and blue (RGB) components (e.g., pixel values) from the graphical model texture. The ideogram component **260** may also determine an alpha value (e.g., a pixel value) from a red channel of the alpha mask. Where the ideogram component **260** determines that

the sticker mask layer will be included in the ideogram, the ideogram component **260** modifies the alpha mask by the sticker mask layer. Where the ideogram component **260** determines the sticker background layer will be included in the ideogram, the ideogram component **260** blends alpha values of a graphical element in the sticker background with that of the alpha mask layer or the graphical model texture. Where the ideogram component **260** determines the sticker foreground layer will be included in the ideogram, the ideogram component **260** blends alpha values of a graphical element in the sticker foreground with the alpha values of the alpha mask layer or the graphical model texture.

(56) FIG. **6** depicts a flow diagram illustrating an example method **600** for generating an ideogram from a set of images of an image stream. The operations of method **600** may be performed by components of the ideogram generation system **160**. In some instances, certain operations of the method **600** may be performed using one or more operations of the method **300** or as sub-operations of one or more operations of the method **300**, as will be explained in more detail below.

(57) In some example embodiments, in response to initiating operation **370**, in operation **610**, the ideogram component **260** determines a target application for the ideogram. The ideogram component **260** may determine the target application based on user interactions with the client device **110**, interactions among applications stored on or currently being processed by at least one processor of the client device **110**, or any other suitable manner. In some instances, the ideogram component **260** determines a target application based on a hand-off initiating the ideogram generation system **160** or the method **300**. For example, a user may interact with a first application presented at the client device **110**. During the interaction with the first application, the user may select a user interface element for generating a new, unique, or tailored ideogram. The first application may initiate a hand-off to the ideogram generation system **160** or one or more components of the ideogram generation system **160**. The target application may be determined as an application which initiates a hand-off to the ideogram generation system **160** for creation of an ideogram.

(58) In some example embodiments, the ideogram component **260** determines the target application for the ideogram based on an application accessing an ideogram library. The ideogram library may contain one or more previously generated ideogram, a previously generated graphical model, and one or more graphical elements for addition to the graphical model in creating ideograms. The ideogram library may be linked to the ideogram component **260** such that accessing the ideogram library causes initiation of operation **610**. For example, the ideogram library may be accessed through the ideogram component **260** by the application. Although described with specified examples, it should be understood that the ideogram component **260** may use any suitable algorithm, method, or set of operations to identify a target application for which an ideogram is to be generated or in which an ideogram is to be used.

(59) In operation **620**, the ideogram component **260** determines one or more dimensions for application ideograms associated with the target application. The one or more dimensions of the application ideograms may be length and width dimensions, diagonal measurement dimensions, pixel measurements (e.g., length, width, or diagonal measurements), pixel counts, or any other suitable dimension. The one or more dimensions for application ideograms may indicate one or more of a minimum size and a maximum size for application ideograms presented within the target application.

(60) In some example embodiments, where an ideogram is being created for use in a target application, the one or more dimensions for application ideograms include position type dimensions. The position type dimensions may represent one or more of a minimum size and a maximum size for graphical elements used in a predetermined position type within an ideogram. The position type may be a foreground position, a background position, and a medial position between the background position and the foreground position. In some instances, the one or more dimensions may include a location within the application ideogram. For example, some foreground

graphical elements may be limited to one or more specified positions within a foreground of an application ideogram and a background graphical element may be limited to one or more specified positions within a background of the application ideogram.

(61) In operation **622**, the ideogram component **260** determines that the position type of a first graphical element is a background type. In some embodiments, based on determining one or more dimensions (e.g., position type dimensions), the ideogram component **260** determines the position types of graphical elements to be included in a generated ideogram. The ideogram component **260** may determine a position type of a graphical element based on identifying a position indication within metadata associated with the graphical element (e.g., the first graphical element). The metadata may indicate whether the graphical element is configured to be positioned in a background, a foreground, or a medial position. In some embodiments, the ideogram component **260** may dynamically determine the position type for a graphical element by matching size, shape, dimensions, or content of the graphical element with size, shape, dimensions, or content characteristics of a specified position type. For example, a background type may have a first predefined size, shape, and dimension characteristic (e.g., a square having the maximum size and dimensions of an application ideogram), while foreground type may have a second predefined size, shape, and dimension characteristic. A graphical element having characteristics matching the first predefined size, shape, and dimension characteristics may be determined to be a background type.

(62) In some instances, where the ideogram component **260** identifies a position type based on content of the graphical element, the ideogram component **260** may identify a content based on metadata for the graphical element, image recognition operations applied to the graphical element, or the title of the graphical element. The ideogram component **260** may match the identified content of the graphical element to a set of content types associated with the background type, the foreground type, or the medial position type. For example, where the graphical element depicts scenery with palm trees and a sunset, the ideogram component **260** may identify the content for the graphical element as scenery. The ideogram component **260** may then parse metadata or description data associated with the background type, the foreground type, and the medial position type to determine which type is associated with scenery. In this example, the ideogram component **260** may identify the scenery graphical element as a background type by determining that the keyword “scenery” is associated with the background type.

(63) In operation **624**, the ideogram component **260** determines that the position type of a second graphical element is a foreground type. The ideogram component **260** may determine the position type of the second graphical element in a manner similar to or the same as described with respect to operation **622**. Although described as determining position types for a first graphical element and a second graphical element having distinct position types, it should be understood that the ideogram component **260** may determine position types for any number of graphical elements and may determine more than one graphical element to be positioned in a single position type.

(64) In operation **630**, the ideogram component **260** generates a modified ideogram based on the one or more dimensions. In embodiments where the ideogram was previously generated, the ideogram component **260** modifies the existing ideogram based on the one or more dimensions identified in operation **620**. The ideogram component **260** may modify the existing ideogram by reducing the dimensions of the existing ideogram to be within the one or more dimensions (e.g., maximum dimensions) identified in operation **620**. In modifying the existing ideogram, the ideogram component **260** may retain an aspect ratio or proportion of the existing ideogram to prevent the existing ideogram from being skewed during the modification. In embodiments where the ideogram is being generated, the ideogram component **260** may modify one or more of the graphical elements and the graphical model included in the ideogram based on the dimensions identified in operation **620**. In these instances, the ideogram component **260** also positions the graphical model and the one or more graphical elements based, at least in part, on the dimensions identified in operation **620**.

(65) In operation **632**, in generating the ideogram, the ideogram component **260** positions the first graphical element, identified as a background type, behind at least a portion of the graphical model. In some instances, the ideogram component **260** positions the first graphical element such that the portion of the graphical model obstructs at least a portion of the graphical element. The ideogram component **260** may also modify a size (e.g., one or more dimensions or measurements) of the first graphical element based on the one or more dimensions identified in operation **620**. The ideogram component **260** may position the first graphical element behind the portion of the graphical model by generating a layered image file and assigning or otherwise placing the first graphical element in a first layer or base layer. The ideogram component **260** may then place the graphical model in a second layer above the first layer such that a portion of the graphical element obscures a portion of the first graphical element.

(66) In operation **634**, the ideogram component **260** positions the graphical model behind the second graphical element such that the graphical element obstructs the portion of one or more of the graphical model and the first graphical element. The ideogram component **260** may also modify a size (e.g., one or more dimensions or measurements) of the second graphical element based on the one or more dimensions identified in operation **620**. The ideogram component **260** may position the second graphical element in front of the portion of the graphical model by applying the second graphical element to the layered image file, assigning or otherwise placing the second graphical element in a third layer above the second layer including the graphical model. In some embodiments, the ideogram component **260** may then flatten or otherwise render the layered image file into the ideogram.

(67) FIG. 7 depicts a flow diagram illustrating an example method **700** for generating an ideogram from a set of images of an image stream. The operations of method **700** may be performed by components of the ideogram generation system **160**. In some instances, certain operations of the method **700** may be performed using one or more operations of the method **300** or as sub-operations of one or more operations of the method **300**, as will be explained in more detail below.

(68) In some example embodiments, as a part of operation **360**, in operation **710**, the ideogram component **260** determines a position type of a graphical element of the one or more graphical elements. The position type may be determined similarly to or the same as the manner described in operations **622** and **624**. The position type may be a foreground position, a background position, or a medial position between the foreground and the background. In some embodiments, operation **710** may be performed in modifying an existing ideogram to include an additional graphical element.

(69) In operation **720**, the ideogram component **260** positions the graphical element based on the position type of the graphical element. The ideogram component **260** may position the graphical element similarly to or the same as the manner described above in operations **632** and **634**. In embodiments where the operations **710** and **720** are performed with respect to an existing ideogram, the ideogram component **260** may position the graphical element in an image layer. The ideogram component **260** may position the graphical element in an existing image layer of an ideogram generated from a layered image file or may generate a new image layer. In some instances, the ideogram component **260** may generate a new layered image file with a first image layer including the existing ideogram. The ideogram component **260** position the graphical element in a second image layer, above or below the first image layer, and generate a new ideogram from the combination of the existing ideogram and the graphical element. The new ideogram may be flattened or otherwise rendered into the new ideogram from the new layered image or the layered image used to generate the existing ideogram.

(70) FIG. 8 depicts a flow diagram illustrating an example method **800** for generating an ideogram from a set of images of an image stream. The operations of method **800** may be performed by components of the ideogram generation system **160**. In some instances, certain operations of the method **800** may be performed using one or more operations of the method **300** or as sub-

operations of one or more operations of the method **300**, as will be explained in more detail below.

(71) In some example embodiments, in operation **810**, the ideogram component **260** renders the graphical model. The graphical model may be a three-dimensional graphical model and the one or more graphical elements may be two-dimensional graphical elements. In instances where the graphical model is a three-dimensional graphical model, in operation **810**, the ideogram component **260** renders the three-dimensional graphical model as a two-dimensional graphical model in response to positioning the one or more graphical elements. The ideogram component **260** may render the three-dimensional graphical model using a flattening process, generating an image file from a depicted view of the three-dimensional graphical model, or any other suitable method.

(72) In operation **820**, the ideogram component **260** renders the ideogram as a two-dimensional ideogram combining the two-dimensional graphical model and the one or more graphical elements. The ideogram component **260** may render the ideogram as a two-dimensional ideogram by generating a layered image and placing each of the one or more graphical elements and the two-dimensional graphical model in layers within the layered image. For example, the ideogram component **260** may assign or otherwise place each of the one or more graphical elements and the two-dimensional graphical model in separate layers within the layered image.

(73) FIG. **9** depicts a flow diagram illustrating an example method **900** for generating an ideogram from a set of images of an image stream. The operations of method **900** may be performed by components of the ideogram generation system **160**. In some instances, certain operations of the method **900** may be performed using one or more operations of the method **300** or as sub-operations of one or more operations of the method **300**, as will be explained in more detail below.

(74) In some example embodiments, as part of operation **370**, in operation **910**, the ideogram component **260** determines a first size for a first graphical element and a second size for a second graphical element. The first size and the second size may correspond to one or more measurements (e.g., a length or height) of the first graphical element and the second graphical element, respectively. In some instances, the sizes determined in operation **910** are a largest measurement of one or more of a length, a height, a diagonal, a circumference, or other suitable measurement.

(75) In operation **920**, the ideogram component **260** determines a prioritized element from the first graphical element and the second graphical element. In some instances, the ideogram component **260** determines the second graphical element is the prioritized element. The prioritized element may be selected based on priority values associated with each of the first graphical element and the second graphical element. For example, each graphical element may include a priority value indicative of a relative importance of the graphical element in ideograms in which the graphical element may be used. The priority value may represent the position type of the graphical element. For example, a background type graphical element may have a relative higher priority than a foreground type graphical element. Where the priority value is tied to the position type for the graphical element, the background type may be presented with a higher priority value as a basis, foundation, or theme on which the ideogram is generated. In some embodiments, the ideogram component **260** determines the prioritized element based on the first size and the second size determined for the first graphical element and the second graphical element, respectively. The ideogram component **260** may determine the prioritized element as a graphical element closest to a maximum size of one or more of the ideogram or a position type without exceeding the maximum size. The ideogram component **260**, in these embodiments, determines the size of each of the first graphical element, the second graphical element, and a corresponding target ideogram size (e.g., a position type size or a maximum ideogram size), and determines which of the first graphical element and the second graphical is the largest and whether one of the graphical elements exceeds the target ideogram size.

(76) In operation **930**, the ideogram component **260** scales one or more of the first graphical element and the second graphical element to generate one or more modified sizes for the first graphical element and the second graphical element. Where the second graphical element is the

prioritized element, the ideogram component **260** may scale the first size of the first graphical element to generate a modified size of the first graphical element based on the second size of the second graphical element. The graphical element being scaled may be resized relative to the other graphical element while retaining its original proportions so as to prevent skewing of the scaled graphical element.

(77) In operation **940**, the ideogram component **260** scales the graphical model (e.g., the composite model) based on the second size of the second graphical element and the modified size of the first graphical element. In some embodiments, the ideogram component **260** scales the graphical model to fit within the ideogram being generated (e.g., fit within a maximum size of the generated ideogram). The graphical model may be scaled to retain original proportions of the graphical model, as well as to fit within a scale of the first graphical element and the second graphical element. For example, where the graphical model is resized and placed in an ideogram having the first graphical element as a block lettered “Hawaii” banner in a foreground and the second graphical as a tropical island in a background, the graphical model may be resized and positioned in a scale and location suitable to appear to stand on a beach of the tropical island with the “Hawaii” banner in front of the graphical model.

(78) FIG. **10** depicts a flow diagram illustrating an example method **1000** for generating an ideogram from a set of images of an image stream. The operations of method **1000** may be performed by components of the ideogram generation system **160**. In some instances, certain operations of the method **1000** may be performed using one or more operations of the method **300** or as sub-operations of one or more operations of the method **300**, as will be explained in more detail below.

(79) In some example embodiments, as part of operation **350**, in operation **1010**, the avatar component **250** generates at least a portion of a body model. The portion of the body model may be connected to the graphical model of the face to generate or form a composite model. The composite model represents all or a portion of a graphical representation of a body. The body model may be generated using a skeletal model movable to position at least a portion of the composite model. Movement of the skeletal model may configure the composite model into poses corresponding to poses of a body. For example, movement of one or more portions of the skeletal model may cause the composite model to appear as a body in a seated position, in a jumping position, waving, or any other suitable body pose.

(80) In operation **1020**, the avatar component **250** determines a pose corresponding to the one or more graphical elements. In some example embodiments, the avatar component **250** determines the pose for the one or more graphical elements based on pose data associated with the one or more graphical elements. The pose data may be included in metadata for the one or more graphical elements and indicate a pose and location at which the composite model is to be placed when an ideogram is generated including the graphical element. For example, where a graphical element is a beach scene including a reclining chair or hammock, the metadata may include position information identifying the location of the reclining chair or hammock within the graphical element and an indication that a seated or reclining pose is appropriate for the reclining chair or hammock. The pose data may include orientation of one or more elements (e.g., arms, legs, hands, feet, or body) of the skeletal model. The orientation may include relative positions of two or more elements of the skeletal model to indicate interaction between the elements, such as crossing arms, crossing legs, hand gestures, arm gestures, or body gestures.

(81) In some example embodiments, the avatar component **250** determines the pose based on prompting user input to identify pose data (e.g., pose and location) for the composite model within the ideogram. The avatar component **250** may generate and cause presentation of one or more user interface elements indicating a set of poses and a set of locations. In these embodiments, the avatar component **250** generates selectable graphical user interface elements including an indication of one or more poses of the set of poses. The indication of a pose may be in the form of a written

description, such as “sitting,” “standing,” “waving,” “jumping,” or other suitable textual descriptions. The indication of the pose may be in the form of a pictographic description, such as a picture of a sitting figure, a standing figure, a waving figure, a reclining figure, a jumping figure, or any other suitable image or animation.

(82) The avatar component **250** may generate selectable graphical user interface elements indicating one or more locations of the set of locations at which the composite model may be placed. The indication of a location may be in the form of a textual description or a pictorial description. The textual descriptions may be provided using a plurality of user interface elements, each having a written description of a position, such as “upper left,” “upper right,” “center,” “seated in chair,” “reclining in hammock,” “jumping from bottom right to bottom left,” or any other suitable textual description. The pictorial descriptions may be in the form of images within user interface elements indicating locations on a background graphical element, a selectable grid positioned on the background graphical element, predetermined identified positions, or any other suitable pictorial indication. Where the pictorial description indicates a predetermined identified position, the predetermined identified position may be shown by a broken line (e.g., cutout) depicting the pose and location of a composite model, a selectable version of the composite model placed in one or more locations within the background, or any other suitable pictorial description.

(83) The graphical user interface elements may be presented on a dedicated graphical user interface presentation (e.g., a rendered screen); as an overlay above a presentation of one or more of the graphical model, the composite model, the one or more graphical elements for use in the ideogram, or a generated or partially generated ideogram; or in any other suitable manner such that the user may be prompted, by selectable elements, to supply user input identifying the pose data. Graphical user interface elements may be presented in an order showing pose elements prior to location elements, where the pose affects the possible locations (e.g., a jumping animation or a reclining pose). In some instances, the graphical user interface elements may be presented to enable a selection of the location prior to selection of the pose or contemporaneous with the selection of the pose.

(84) The avatar component **250** may also generate and cause presentation of the one or more user interface elements indicating the set of poses and enabling free form selection of the location of the composite model within the ideogram. In these example embodiments, the avatar component **250** may generate and cause presentation of user interface elements for pose selection as described above. Free form selection of the location may be enabled by generating and causing presentation of a draggable or otherwise positionable version of the composite model. The positionable composite model may be presented within or over a background graphical element to be used as a background for the ideogram. In some example embodiments, the composite model may be presented in a selected pose. The positionable composite model may also be presented in a neutral pose (e.g., a pose which has not previously been selected), and the avatar component **250** may enable selection of the pose after the composite model is positioned within the background graphical element.

(85) In operation **1030**, the avatar component **250** positions one or more portions of the skeletal model of the composite model to represent the pose of operation **1020**. The avatar component **250** may position the skeletal model of the composite model based on one or more of the pose data for the one or more graphical elements and the user input indicating at least one of a pose and a location. The avatar component **250** may position the one or more portions of the skeletal model by matching positions or orientations indicated in the pose data or the user input, placing the graphical model (e.g., the face model) on a pre-generated body model matching the pose, or any other suitable manner.

(86) In operation **1040**, the ideogram component **260** generates the ideogram with the one or more graphical element and the composite model positioned in the pose. The ideogram component **260** may generate the ideogram similarly to or the same as described above with respect to operations

(87) Modules, Components, and Logic

(88) Certain embodiments are described herein as including logic or a number of components, modules, or mechanisms. Components can constitute hardware components. A “hardware component” is a tangible unit capable of performing certain operations and can be configured or arranged in a certain physical manner. In various example embodiments, computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or hardware components of a computer system (e.g., at least one hardware processor, a processor, or a group of processors) is configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

(89) In some embodiments, a hardware component is implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component can include dedicated circuitry or logic that is permanently configured to perform certain operations. For example, a hardware component can be a special-purpose processor, such as a Field-Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component can include software encompassed within a general-purpose processor or other programmable processor. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) can be driven by cost and time considerations.

(90) Accordingly, the phrase “hardware component” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. As used herein, “hardware-implemented component” refers to a hardware component. Considering embodiments in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software can accordingly configure a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time.

(91) Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components can be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications can be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In embodiments in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component performs an operation and stores the output of that operation in a memory device to which it is communicatively coupled. A further hardware component can then, at a later time, access the memory device to retrieve and process the stored output. Hardware components can also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

(92) The various operations of example methods described herein can be performed, at least partially, by processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured,

such processors constitute processor-implemented components that operate to perform operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using processors.

(93) Similarly, the methods described herein can be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method can be performed by processors or processor-implemented components. Moreover, the processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via appropriate interfaces (e.g., an Application Program Interface (API)).

(94) The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented components are located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented components are distributed across a number of geographic locations.

(95) Applications

(96) FIG. 11 illustrates an example mobile device **1100** executing a mobile operating system (e.g., IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems), consistent with some embodiments. In one embodiment, the mobile device **1100** includes a touch screen operable to receive tactile data from a user **1102**. For instance, the user **1102** may physically touch **1104** the mobile device **1100**, and in response to the touch **1104**, the mobile device **1100** may determine tactile data such as touch location, touch force, or gesture motion. In various example embodiments, the mobile device **1100** displays a home screen **1106** (e.g., Springboard on IOS™) operable to launch applications or otherwise manage various aspects of the mobile device **1100**. In some example embodiments, the home screen **1106** provides status information such as battery life, connectivity, or other hardware statuses. The user **1102** can activate user interface elements by touching an area occupied by a respective user interface element. In this manner, the user **1102** interacts with the applications of the mobile device **1100**. For example, touching the area occupied by a particular icon included in the home screen **1106** causes launching of an application corresponding to the particular icon.

(97) The mobile device **1100**, as shown in FIG. 11, includes an imaging device **1108**. The imaging device **1108** may be a camera or any other device coupled to the mobile device **1100** capable of capturing a video stream or one or more successive images. The imaging device **1108** may be triggered by the ideogram generation system **160** or a selectable user interface element to initiate capture of a video stream or succession of images and pass the video stream or succession of images to the ideogram generation system **160** for processing according to the one or more methods described in the present disclosure.

(98) Many varieties of applications (also referred to as “apps”) can be executing on the mobile device **1100**, such as native applications (e.g., applications programmed in Objective-C, Swift, or another suitable language running on IOS™, or applications programmed in Java running on ANDROID™), mobile web applications (e.g., applications written in Hypertext Markup Language-5 (HTML5)), or hybrid applications (e.g., a native shell application that launches an HTML5 session). For example, the mobile device **1100** includes a messaging app, an audio recording app, a camera app, a book reader app, a media app, a fitness app, a file management app, a location app, a browser app, a settings app, a contacts app, a telephone call app, or other apps (e.g., gaming apps, social networking apps, biometric monitoring apps). In another example, the mobile device **1100** includes a social messaging app **1110** such as SNAPCHAT® that, consistent with some embodiments, allows users to exchange ephemeral messages that include media content. In this

example, the social messaging app **1110** can incorporate aspects of embodiments described herein. For example, in some embodiments the social messaging application includes an ephemeral gallery of media created by users the social messaging application. These galleries may consist of videos or pictures posted by a user and made viewable by contacts (e.g., “friends”) of the user. Alternatively, public galleries may be created by administrators of the social messaging application consisting of media from any users of the application (and accessible by all users). In yet another embodiment, the social messaging application may include a “magazine” feature which consists of articles and other content generated by publishers on the social messaging application's platform and accessible by any users. Any of these environments or platforms may be used to implement concepts of the present inventive subject matter.

(99) In some embodiments, an ephemeral message system may include messages having ephemeral video clips or images which are deleted following a deletion trigger event such as a viewing time or viewing completion. In such embodiments, a device implementing the ideogram generation system **160** may identify, track, extract, and modify an area of interest and the color depicted therein within the ephemeral video clip, as the ephemeral video clip is being captured by the device, and transmit the ephemeral video clip to another device using the ephemeral message system.

(100) Software Architecture

(101) FIG. **12** is a block diagram **1200** illustrating an architecture of software **1202**, which can be installed on the devices described above. FIG. **12** is merely a non-limiting example of a software architecture, and it will be appreciated that many other architectures can be implemented to facilitate the functionality described herein. In various embodiments, the software **1202** is implemented by hardware such as machine **1300** of FIG. **13** that includes processors **1310**, memory **1330**, and I/O components **1350**. In this example architecture, the software **1202** can be conceptualized as a stack of layers where each layer may provide a particular functionality. For example, the software **1202** includes layers such as an operating system **1204**, libraries **1206**, frameworks **1208**, and applications **1210**. Operationally, the applications **1210** invoke application programming interface (API) calls **1212** through the software stack and receive messages **1214** in response to the API calls **1212**, consistent with some embodiments.

(102) In various implementations, the operating system **1204** manages hardware resources and provides common services. The operating system **1204** includes, for example, a kernel **1220**, services **1222**, and drivers **1224**. The kernel **1220** acts as an abstraction layer between the hardware and the other software layers consistent with some embodiments. For example, the kernel **1220** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services **1222** can provide other common services for the other software layers. The drivers **1224** are responsible for controlling or interfacing with the underlying hardware, according to some embodiments. For instance, the drivers **1224** can include display drivers, camera drivers, BLUETOOTH® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

(103) In some embodiments, the libraries **1206** provide a low-level common infrastructure utilized by the applications **1210**. The libraries **1206** can include system libraries **1230** (e.g., C standard library) that can provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1206** can include API libraries **1232** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g.,

WebKit to provide web browsing functionality), and the like. The libraries **1206** can also include a wide variety of other libraries **1234** to provide many other APIs to the applications **1210**.

(104) The frameworks **1208** provide a high-level common infrastructure that can be utilized by the applications **1210**, according to some embodiments. For example, the frameworks **1208** provide various graphic user interface (GUI) functions, high-level resource management, high-level location services, and so forth. The frameworks **1208** can provide a broad spectrum of other APIs that can be utilized by the applications **1210**, some of which may be specific to a particular operating system or platform.

(105) In an example embodiment, the applications **1210** include a home application **1250**, a contacts application **1252**, a browser application **1254**, a book reader application **1256**, a location application **1258**, a media application **1260**, a messaging application **1262**, a game application **1264**, and a broad assortment of other applications such as a third party application **1266**.

According to some embodiments, the applications **1210** are programs that execute functions defined in the programs. Various programming languages can be employed to create the applications **1210**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third party application **1266** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® PHONE, or another mobile operating systems. In this example, the third party application **1266** can invoke the API calls **1212** provided by the operating system **1204** to facilitate functionality described herein.

Example Machine Architecture and Machine-Readable Medium

(106) FIG. **13** is a block diagram illustrating components of a machine **1300**, according to some embodiments, able to read instructions (e.g., processor executable instructions) from a machine-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any of the methodologies discussed herein. Specifically, FIG. **13** shows a diagrammatic representation of the machine **1300** in the example form of a computer system, within which instructions **1316** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1300** to perform any of the methodologies discussed herein can be executed. In alternative embodiments, the machine **1300** operates as a standalone device or can be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1300** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1300** can comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1316**, sequentially or otherwise, that specify actions to be taken by the machine **1300**. Further, while only a single machine **1300** is illustrated, the term “machine” shall also be taken to include a collection of machines **1300** that individually or jointly execute the instructions **1316** to perform any of the methodologies discussed herein.

(107) In various embodiments, the machine **1300** comprises processors **1310**, memory **1330**, and I/O components **1350**, which can be configured to communicate with each other via a bus **1302**. In an example embodiment, the processors **1310** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) include, for example, a processor **1312** and a

processor **1314** that may execute the instructions **1316**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (also referred to as “cores”) that can execute instructions **1316** contemporaneously. Although FIG. **13** shows multiple processors **1310**, the machine **1300** may include a single processor with a single core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

(108) The memory **1330** comprises a main memory **1332**, a static memory **1334**, and a storage unit **1336** accessible to the processors **1310** via the bus **1302**, according to some embodiments. The storage unit **1336** can include a machine-readable medium **1338** on which are stored the instructions **1316** embodying any of the methodologies or functions described herein. The instructions **1316** can also reside, completely or at least partially, within the main memory **1332**, within the static memory **1334**, within at least one of the processors **1310** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1300**. Accordingly, in various embodiments, the main memory **1332**, the static memory **1334**, and the processors **1310** are considered machine-readable media **1338**.

(109) As used herein, the term “memory” refers to a machine-readable medium **1338** able to store data temporarily or permanently and may be taken to include, but not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, and cache memory. While the machine-readable medium **1338** is shown in an example embodiment to be a single medium, the term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store the instructions **1316**. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions (e.g., instructions **1316**) for execution by a machine (e.g., machine **1300**), such that the instructions, when executed by processors of the machine **1300** (e.g., processors **1310**), cause the machine **1300** to perform any of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” shall accordingly be taken to include, but not be limited to, data repositories in the form of a solid-state memory (e.g., flash memory), an optical medium, a magnetic medium, other non-volatile memory (e.g., Erasable Programmable Read-Only Memory (EPROM)), or any suitable combination thereof. The term “machine-readable medium” specifically excludes non-statutory signals per se.

(110) The I/O components **1350** include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. In general, it will be appreciated that the I/O components **1350** can include many other components that are not shown in FIG. **13**. The I/O components **1350** are grouped according to functionality merely for simplifying the following discussion, and the grouping is in no way limiting. In various example embodiments, the I/O components **1350** include output components **1352** and input components **1354**. The output components **1352** include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components **1354** include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(111) In some further example embodiments, the I/O components **1350** include biometric

components **1356**, motion components **1358**, environmental components **1360**, or position components **1362**, among a wide array of other components. For example, the biometric components **1356** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or mouth gestures), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1358** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environmental components **1360** include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensor components (e.g., machine olfaction detection sensors, gas detection sensors to detect concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1362** include location sensor components (e.g., a Global Positioning System (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(112) Communication can be implemented using a wide variety of technologies. The I/O components **1350** may include communication components **1364** operable to couple the machine **1300** to a network **1380** or devices **1370** via a coupling **1382** and a coupling **1372**, respectively. For example, the communication components **1364** include a network interface component or another suitable device to interface with the network **1380**. In further examples, communication components **1364** include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, BLUETOOTH® components (e.g., BLUETOOTH® Low Energy), WI-FI components, and other communication components to provide communication via other modalities. The devices **1370** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a Universal Serial Bus (USB)).

(113) Moreover, in some embodiments, the communication components **1364** detect identifiers or include components operable to detect identifiers. For example, the communication components **1364** include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect a one-dimensional bar codes such as a Universal Product Code (UPC) bar code, multi-dimensional bar codes such as a Quick Response (QR) code, Aztec Code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, Uniform Commercial Code Reduced Space Symbology (UCC RSS)-2D bar codes, and other optical codes), acoustic detection components (e.g., microphones to identify tagged audio signals), or any suitable combination thereof. In addition, a variety of information can be derived via the communication components **1364**, such as location via Internet Protocol (IP) geo-location, location via WI-FI® signal triangulation, location via detecting a BLUETOOTH® or NFC beacon signal that may indicate a particular location, and so forth.

(114) Transmission Medium

(115) In various example embodiments, portions of the network **1380** can be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a WI-FI® network, another type of network, or a combination of two or more such

networks. For example, the network **1380** or a portion of the network **1380** may include a wireless or cellular network, and the coupling **1382** may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or another type of cellular or wireless coupling. In this example, the coupling **1382** can implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long range protocols, or other data transfer technology.

(116) In example embodiments, the instructions **1316** are transmitted or received over the network **1380** using a transmission medium via a network interface device (e.g., a network interface component included in the communication components **1364**) and utilizing any one of a number of well-known transfer protocols (e.g., Hypertext Transfer Protocol (HTTP)). Similarly, in other example embodiments, the instructions **1316** are transmitted or received using a transmission medium via the coupling **1372** (e.g., a peer-to-peer coupling) to the devices **1370**. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying the instructions **1316** for execution by the machine **1300**, and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

(117) Furthermore, the machine-readable medium **1338** is non-transitory (in other words, not having any transitory signals) in that it does not embody a propagating signal. However, labeling the machine-readable medium **1338** “non-transitory” should not be construed to mean that the medium is incapable of movement; the medium should be considered as being transportable from one physical location to another. Additionally, since the machine-readable medium **1338** is tangible, the medium may be considered to be a machine-readable device.

(118) Language

(119) Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of methods are illustrated and described as separate operations, individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

(120) Although an overview of the inventive subject matter has been described with reference to specific example embodiments, various modifications and changes may be made to these embodiments without departing from the broader scope of embodiments of the present disclosure. Such embodiments of the inventive subject matter may be referred to herein, individually or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single disclosure or inventive concept if more than one is, in fact, disclosed.

(121) The embodiments illustrated herein are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed. Other embodiments may be used and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. The Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

(122) As used herein, the term “or” may be construed in either an inclusive or exclusive sense. Moreover, plural instances may be provided for resources, operations, or structures described herein as a single instance. Additionally, boundaries between various resources, operations, components, engines, and data stores are somewhat arbitrary, and particular operations are illustrated in a context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within a scope of various embodiments of the present disclosure. In general, structures and functionality presented as separate resources in the example configurations may be implemented as a combined structure or resource. Similarly, structures and functionality presented as a single resource may be implemented as separate resources. These and other variations, modifications, additions, and improvements fall within a scope of embodiments of the present disclosure as represented by the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A method comprising: capturing, by an image capture device, a video comprising an image; identifying a face within the image, the face including a plurality of facial features; generating a graphical model of the identified face; identifying position selections indicating placement of a first graphical element and a second graphical element in the image from a plurality of graphical elements, wherein the plurality of graphical elements comprise pose elements and location elements; positioning the first and second graphical elements on the graphical model corresponding to the position selections on the image to generate a media content item; determining maximum dimension information for a target application; based on a comparison between sizes of the first and second graphical elements with the maximum dimension information, determine that the second graphical element is a prioritized element; scaling a first size of the first graphical element based on a second size of the second graphical element, the scaling being based on a sampling of the original first graphical element to generate a scaled size of the first graphical element; scaling the graphical model based on the second size of the second graphical element and the scaled size of first graphical element; generating the media content item using based on the scaled graphical model and the pose elements of the face within the image; and transmitting the generated media content item to the target application.
2. The method of claim 1, further comprising: displaying, together with the image depicting the face in a first region of the display, the graphical model in a second region of the display; generating a digital sticker that includes the graphical model and the first graphical element and a second graphical element, the first graphical element having a first size and the second graphical element having a second size, wherein the first graphical element is placed at the given position with respect to the graphical model based on the identified position selections; determining the second size of the second graphical element in relation to a minimum dimension limit for digital stickers in a messaging application; scaling the second size of the second graphical element relative to the first size of the first graphical element to fit within the minimum dimension limit in response to determining the second size of the second graphical element in relation to the minimum dimension limit; scaling the graphical model to generate a scaled graphical model based on the first size of the first graphical element and the scaled second size of the second graphical element; and causing the digital sticker to be displayed within the messaging application.
3. The method of claim 1, further comprising: displaying a plurality of graphical elements in an order, wherein the order in which the plurality of graphical elements is displayed comprises displaying the pose elements prior to the location elements, wherein selection of a given pose element modifies the location elements, wherein the first graphical element is selected by the input from the displayed plurality of graphical elements; identifying a set of facial landmarks within the portion of the face depicted within the image; identifying expected but missing facial landmarks

within the portion of the face; in response to identifying the set of facial landmarks and in response to identifying the expected but missing facial landmarks, determining one or more characteristics representing the portion of the face depicted in the image; rendering a base face and applying one or more generated features corresponding to the one or more characteristics and the set of facial landmarks to generate the graphical model; and positioning first graphical element proximate to the graphical model of the face.

4. The method of claim 3, wherein determining the one or more characteristics comprises determining color of one or more features depicted on the face to update the graphical model, and wherein positioning the first graphical element comprises: receiving a video comprising a plurality of frames; determining that the set of facial landmarks appear in a first set of the frames and do not appear in a second set of the frames, wherein the set of facial landmarks are identified in response to determining that the set of facial landmarks appear in the first set of the frames; determining that a position type of the first graphical element is a background type; and positioning the first graphical element behind at least a portion of the graphical model such that the portion of the graphical model obstructs at least a portion of the first graphical element.

5. The method of claim 3, wherein determining the one or more characteristics comprises determining a relative color between an area of the portion of the face and one or more features depicted on the portion of the face to update the graphical model, and wherein positioning the first graphical elements comprises: generating the graphical model comprising an avatar based on an image captured with a camera in response to receiving input that selects a generate sticker button; in response to generating the graphical model comprising the avatar, displaying a plurality of options comprising a save avatar option, a customize avatar option, and a generate ideogram option; determining that a position type of a second graphical element is a foreground type; and positioning at least a portion of the graphical model behind the second graphical element such that the second graphical element obstructs the portion of the graphical model.

6. The method of claim 1, wherein the graphical model is a three-dimensional graphical model and the first graphical element is a two-dimensional graphical element, further comprising: rendering the three-dimensional graphical model as a two-dimensional graphical model; rendering a digital sticker as a two-dimensional digital sticker by combining the two-dimensional graphical model and the one or more graphical elements; and moving the first graphical element between one or more predetermined positions.

7. The method of claim 1 further comprising: determining that second graphical element is of a background type and the first graphical element is of a foreground type; and based on determining that second graphical element is of the background type and the first graphical element is of the foreground type, determining the second graphical element is a prioritized element.

8. The method of claim 1, further comprising: receiving a user selection of a theme associated with a digital sticker; and selecting, in response to receiving the user selection of the theme, the first graphical element from a plurality of graphical elements to generate the digital sticker, the first graphical element being associated with the theme.

9. The method of claim 1, further comprising: generating at least a portion of a body model connected to the graphical model of the face to generate a composite model, the body model having a skeletal representation movable to position at least a portion of the composite model; determining a pose corresponding to the first graphical element; positioning one or more portions of the skeletal representation of the composite model to represent the pose; and generating a digital sticker with the first graphical element and the composite model positioned in the pose.

10. The method of claim 1, further comprising: receiving a user selection of a destination application from a plurality of destination applications, the selected destination application comprising a messaging application; in response to receiving the user selection of the destination application, obtaining configuration information for the destination application comprising dimensions and formatting specifications; and generating a digital sticker or avatar to be compliant

with the dimensions and formatting specifications of the destination application.

11. The method of claim 1, further comprising: sending a message comprising a digital sticker to a recipient, the message remaining accessible to the recipient for a predefined duration specified by a sender of the message, the predefined duration beginning when the message is first accessed by the recipient, wherein the message is automatically deleted after the predefined duration elapses.

12. The method of claim 1, wherein the plurality of graphical elements comprises an ordered presentation of graphical elements in which pose elements are presented prior to location elements.

13. The method of claim 1, wherein the first graphical element comprises an animation that is presented together with the graphical model representing the portion of the face.

14. The method of claim 1, further comprising identifying content of the graphical element based on a title of the graphical element, wherein a position type is identified based on the identified content of the graphical element.

15. The method of claim 1, wherein the plurality of facial features include eyes, nose and a mouth, further comprising generating a digital sticker using universal configuration information to enable the digital sticker to be used across a plurality of destination applications.

16. The method of claim 1, further comprising: displaying the video in a first region of a display, the graphical model generated during display of the video; and displaying the graphical model concurrently with the video in a second region, the identified position selections via user input onto the first region, and the second region updated based on the user position selections, wherein the first region of the display includes a complete representation of the face comprising a plurality of features, and wherein a second region includes a complete graphical model that graphically represents each of the plurality of features of the face, the first region being displayed side-by-side with the second region.

17. The method of claim 1, further comprising: determining a minimum size for the media content item based on one or more characteristics of the target application, wherein scaling the size of the first graphical element is based on both the minimum size for the media content item and the maximum dimension information for the target application.

18. The method of claim 1, wherein the scaling being based on the sampling of the original first graphical element is performed by subsampling the first graphical element.

19. The method of claim 1, wherein the scaling being based on the sampling of the original first graphical element is performed by downsampling the first graphical element.

20. The method of claim 1, wherein scaling the first size of the first graphical element based on the second size of the second graphical element comprises scaling the first size of the first graphical element relative to the second size of the second graphical element.

21. A system comprising: one or more processors configured to perform operations comprising: capturing, by an image capture device, a video comprising an image; identifying a face within the image, the face including a plurality of facial features; generating a graphical model of the identified face; identifying position selections indicating placement of a first graphical element and a second graphical element in the image from a plurality of graphical elements, wherein the plurality of graphical elements comprise pose elements and location elements; positioning the first and second graphical elements on the graphical model corresponding to the position selections on the image to generate a media content item; determining maximum dimension information for a target application; based on a comparison between sizes of the first and second graphical elements with the maximum dimension information, determine that the second graphical element is a prioritized element; scaling a first size of the first graphical element based on a second size of the second graphical element, the scaling being based on a sampling of the original first graphical element to generate a scaled size of the first graphical element; scaling the graphical model based on the second size of the second graphical element and the scaled size of first graphical element; generating the media content item using based on the scaled graphical model and the pose of the face within the image; and transmitting the generated media content item to the target application.

22. A non-transitory processor-readable storage medium storing processor executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: capturing, by an image capture device, a video comprising an image; identifying a face within the image, the face including a plurality of facial features; generating a graphical model of the identified face; identifying position selections indicating placement of a first graphical element and a second graphical element in the image from a plurality of graphical elements, wherein the plurality of graphical elements comprise pose elements and location elements; positioning the first and second graphical elements on the graphical model corresponding to the position selections on the image to generate a media content item; determining maximum dimension information for a target application; based on a comparison between sizes of the first and second graphical elements with the maximum dimension information, determine that the second graphical element is a prioritized element; scaling a first size of the first graphical element based on a second size of the second graphical element, the scaling being based on a sampling of the original first graphical element to generate a scaled size of the first graphical element; scaling the graphical model based on the second size of the second graphical element and the scaled size of first graphical element; generating the media content item using based on the scaled graphical model and the pose of the face within the image; and transmitting the generated media content item to the target application.
